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Disagreement of disagreement

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ABSTRACT

We show that leading investor disagreement proxies have near-zero cross-correlation – making disagreement pricing inferences challenging. We develop a frictionless model that generates overpricing from latent disagreement, jointly rationalizes major proxies, and generates new testable predictions. Equilibrium interrelationships motivate a new disagreement measure, DIS, which is more predictive of returns cross-sectionally than major proxies and more consistently predictive across macroeconomic regimes. We provide evidence that the friction-based channel of Miller (1977) and a frictionless mechanism coexist and are empirically separable. DIS also provides a micro-founded, disagreement-based account of the IVOL puzzle, absorbing IVOL's predictability while revealing its regime and friction dependence.

1. Introduction

Disagreement among investors is ubiquitous in financial markets but fundamentally unobservable. The literature has responded by proposing a wide array of proxies, each with theoretical motivation and empirical support in some settings. Yet, the proxies disagree substantially with each other. Average pairwise correlations are close to zero among the three major categories of firm-level disagreement measures: analyst forecast dispersion, idiosyncratic volatility (IVOL), and short interest volume.[1] This divergence means that empirical findings about the relationship between disagreement and asset returns depend heavily on which proxy is chosen, contradictory results across samples are a natural consequence of proxy selection, and no existing measure can claim to capture the latent disagreement variable rather than one of its observable projections. For example, volume-based measures could also proxy for liquidity (Benston and Hagerman, 1974; Petersen and Fialkowski, 1994) and volatility-based measures such as IVOL could also proxy for risk (Goyal and Santa-Clara, 2003) or firm transparency (Ferreira and Laux, 2007). We call this the *disagreement of disagreement* problem, and we offer a resolution.

Our resolution proceeds in three steps: (i) a theoretical framework that links all major proxies to a single latent disagreement variable, explains why existing disagreement proxies disagree with one another, and provides testable predictions; (ii) a composite measure guided by the framework's equilibrium interrelationships to produce an empirical vehicle to test those predictions; and (iii) an empirical implementation that is consistent with the framework's predictions. Our analyses provide new evidence on the nature of investor disagreement as a mispricing phenomenon with its model-implied structure as essential to its empirical performance, and shed new light on related unresolved questions including a potential micro-founded resolution of the IVOL puzzle.

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[1] These three categories are among the most common firm-specific disagreement measures in the literature and are based on the least stale information for a broad cross-section of stocks (Huang et al., 2021). Boehme et al. (2006) and Berkman et al. (2009) build on prior theory relating belief heterogeneity to volatility to propose IVOL as a firm-level measure of disagreement. See Internet Appendix E for a summary of disagreement proxy families employed across accounting, economics, finance, and management literatures, and Goulding et al. (2026) for detailed discussion and additional background on each family.

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The theoretical framework. We analyze a simple two-period market for one risky asset in which the equilibrium price is determined endogenously by a standard market-clearing approach across a continuum of expected-utility-maximizing investors with differences of opinion.[2] We generalize this approach to a multi-asset framework under a factor structure in Appendix B, in which the common factor structure has mutually independent idiosyncratic components and disagreement is about idiosyncratic expected payoffs, grounding our single-asset focus for cross-sectional predictions.

The framework imposes no short-sale frictions and no parametric assumptions on the asset's payoff distribution. Standard utility assumptions apply: investors prefer more to less, are risk averse, and have demand for precautionary savings when facing uncertainty or positive intensity of prudence (Kimball, 1990). The key mechanism is that when asset payoffs are positively skewed, as they are for the vast majority of stocks (Boyer et al., 2010), optimistic investors buy more aggressively than pessimists sell short. This asymmetry arises because a short position in a positively skewed asset is equivalent to a long position in a negatively skewed one, which under standard skewness preferences commands lower-magnitude demand. When investors also disagree, this demand asymmetry means that the market-clearing price is pushed above what fundamentals warrant, and more disagreement implies greater overpricing (mispricing) and lower subsequent expected returns. The model identifies the ratio of absolute prudence to absolute risk aversion as the coefficient governing the magnitude of the disagreement effect on expected returns, a distinction that collapses under the CARA utility standard but is operative under more general preferences including CRRA (power) utility, where it enters as a common scalar across assets.

The mechanism does not require short-sale frictions to operate, unlike the leading explanation in the literature following Miller (1977). We demonstrate that the combination of heterogeneous beliefs and positive asset skewness is sufficient to produce overpricing in a frictionless market, though frictions, where present, may provide additional amplification. This theoretical result motivates our empirical investigation of whether the disagreement–return relationship persists in the cross section among stocks where short-selling constraints are least binding (supporting evidence in Section 5.5).

We model investor beliefs via two components: the relative distribution of beliefs across investors and the scale of those beliefs. This separation accommodates many possible shapes of heterogeneity, such as polarized, uniform, concentrated-with-tails, and asymmetric, while treating the overall level of disagreement as a separate, cleanly defined parameter. The model formally distinguishes analyst earnings forecast dispersion from investor disagreement. Forecast dispersion amplifies the scale of existing belief heterogeneity rather than creating it, a distinction reflected in our approach and overlooked by prior work that treats analyst dispersion as a direct disagreement substitute (Rees and Thomas, 2010; Dittmar and Thakor, 2007; Park, 2005; cf. Goulding, 2017, 2018). In our framework, dispersion and idiosyncratic disagreement are correlated but separate constructs, and the composite measure reflects their interaction rather than treating analyst dispersion as a standalone proxy.

The framework generates additional predictions that do not arise from prior models of disagreement and skewness: a correctly specified idiosyncratic disagreement measure should exhibit cross-sectional return predictability that is stable across macroeconomic regimes (from the multi-asset separation); and the specific nonlinear functional form of the composite measure discussed below is essential and not recoverable by linear alternatives. These predictions are tested in Section 5 and the evidence is consistent with our framework.

The composite measure. The model's equilibrium conditions relate the unobservable scale of disagreement to observable asset-specific variables: idiosyncratic volatility, short interest, analyst forecast dispersion, idiosyncratic skewness, and uncertainty about the mean payoff. By solving for disagreement as a function of these variables, we derive a composite disagreement measure, DIS, which is a nonlinear function of observable components, several of which correspond to common proxies of disagreement in the literature. This nonlinearity reflects the equilibrium interplay and implicit cross-validation among proxies.

Empirical findings. We employ DIS as the vehicle through which we empirically examine our theoretical framework and its structural predictions. Our sample spans January 1994 to December 2022, restricted to stocks followed by at least two analysts, a restriction that excludes the smallest, most illiquid, and most retail-dominated stocks where anomaly profits are typically concentrated. Our empirical analyses are organized into eight sets of interlocking tests of a single framework, providing mutually reinforcing cumulative evidence.

First, a value-weighted decile spread portfolio sorted by DIS produces 17%–21% annualized alpha (dependent on the factor model), substantially exceeding the alphas produced by any individual disagreement proxy on the same sample. Alphas are driven primarily by the short leg, which the model predicts directly. Demand convexity from positive skewness amplifies disagreement's overpricing effects at the firm level, producing short-leg dominance independently of aggregate sentiment or probability weighting. The framework also explains why these alphas should survive standard risk adjustments (Appendix B.3) and why they persist. The same skewness preferences that generate overpricing discourage the corrective arbitrage trades, providing a preference-based complement to friction-based limits to arbitrage (Shleifer and Vishny, 1997; Pontiff, 2006). Alphas are robust to further exclusion of the most illiquid and retail-owned stocks from our already relatively liquid sample (Section 5.1).

Second, in Fama and MacBeth (1973) regressions, DIS generates substantial return predictability in the cross section – almost four standard errors from zero. This result is robust to controls for other disagreement measures and standard cross-sectional predictors.

Third, DIS provides a potential disagreement-based solution to the idiosyncratic volatility puzzle. Ang et al. (2006) document that stocks with high idiosyncratic volatility earn anomalously low subsequent returns, a finding that Hou and Loh (2016) characterize, after an exhaustive survey, as unresolved. IVOL is, in our model, a closely linked individual proxy to latent disagreement, inheriting negative return predictability that latent disagreement generates. Empirically, IVOL exhibits the highest correlation with DIS among proxies, but DIS absorbs IVOL's predictability in Fama and MacBeth (1973) regressions. IVOL's coefficient becomes statistically insignificant once DIS is included and this absorption holds across nearly all uncertainty regimes and in subsamples of smaller stocks (among those covered by two or more analysts), where IVOL's standalone predictability is strongest.

Fourth, DIS behaves as the model predicts for an idiosyncratic disagreement measure rather than one contaminated by aggregate uncertainty. Our framework implies a testable restriction: a well-specified idiosyncratic disagreement measure should predict returns independently of aggregate macro conditions, because aggregate variation is absorbed into factor premia. We test this restriction using eight diverse aggregate indicators spanning market stress, aggregate uncertainty, and investor sentiment. DIS predicts returns consistently in both high and low states across all eight indicators, while the individual proxies of IVOL and short interest predict returns significantly primarily in low-uncertainty environments. These patterns offer a structural account of the conflicting results across samples using different proxies, each

[2] Our model adopts the differences-of-opinion paradigm in which price influences investor trades but not beliefs; see Varian (1985), Harris and Raviv (1993), Shalen (1993), Kandel and Pearson (1995), and Cao and Ou-Yang (2009). Differences of opinion are consistent with investors not conditioning their decisions on prices (Banerjee, 2011; Banerjee et al., 2024). Alternatively, insensitivity of beliefs to price can be approximately true under slow information diffusion (Hong and Stein, 1999), or with the addition of noise traders (e.g., Grossman and Stiglitz, 1980; Diamond and Verrecchia, 1981).

capturing a distinct and partial slice of latent disagreement whose predictive content varies with the macro environment. More broadly, regime-dependence of predictive power can serve as a diagnostic for aggregate contamination of idiosyncratic proxies, a principle that extends beyond the disagreement setting.

Fifth, DIS provides evidence on a question debated largely on theoretical grounds since Miller (1977) of whether disagreement can impact pricing without short-sale frictions. Using institutional ownership as an inverse proxy for short-selling frictions following Nagel (2005), we find that both channels are operative and empirically separable. IVOL and short interest exhibit return predictability concentrated where frictions are more binding, as the Miller (1977) channel predicts, whereas DIS exhibits significant and stable return predictability even as institutional ownership approaches 100%, with an interaction with frictions indistinguishable from zero. These contrasting patterns, obtained within a common specification on the same sample, provide what is to our knowledge the first empirical evidence that a frictionless disagreement channel operates alongside the friction-based channel.

Sixth, the model-implied nonlinear structure of DIS is essential to its performance. In Fama and MacBeth (1973) regressions, the return predictability embedded in DIS is not recoverable by linear combinations of the same underlying information. Neither equally weighted composites nor principal components of the same or other common disagreement proxies carry marginal predictive power once DIS is included.

Seventh, we show that DIS detects economically meaningful disagreement that is undetected by individual proxies. We report the characteristics typical of disagreement that goes undetected, underscoring that DIS isolates the latent variable more completely than standalone measures.

Finally, we document the firm-level characteristics associated with elevated disagreement and find that our model-implied measure aligns intuitively with the expected relationship between firm opacity and investor disagreement. For example, higher-DIS firms tend to exhibit increased financial constraints, younger age, intensified R&D expenditures, and reduced volumes of Google searches, both overall and specifically investment-related (see Internet Appendix I).

Relationship to the literature. A large literature studies the role of differing opinions in asset pricing (see Chen et al., 2002; Diether et al., 2002; Yu, 2011; Carlin et al., 2014; Fischer et al., 2022; and the surveys of Curcuru et al., 2010, and Scheinkman and Xiong, 2004). Key strands include investor heterogeneity as a risk factor (Barry and Brown, 1985; Merton, 1987; Varian, 1989); static models of short-sale constraints and disagreement (Miller, 1977; Jarrow, 1980; Diamond and Verrecchia, 1987; Chen et al., 2002); dynamic models and speculative bubbles (Harrison and Kreps, 1978; Scheinkman and Xiong, 2003; Hong et al., 2006); heterogeneous beliefs as a driver of business cycles and asset prices (Bigio et al., 2026); alternative channels via illiquidity (Sadka and Scherbina, 2007), information risk (Johnson, 2004), financial distress (Avramov et al., 2009), and rational learning (Goulding, 2017, 2018; Albagli et al., 2024); connections with trading volume, volatility, and informed trading (Harris and Raviv, 1993; Kandel and Pearson, 1995; Cao and Ou-Yang, 2009; Dumas et al., 2009; Banerjee and Kremer, 2010; Ottaviani and Sørensen, 2015; Atmaz and Basak, 2018; Banerjee et al., 2018; Chabakauri and Han, 2020; see Hong and Stein, 2007; Xiong, 2013; and Goulding et al., 2026 for surveys). Our paper differs in that our results do not rely on firm characteristics, market frictions, or parametric assumptions on the payoff distribution, and we establish linkages among different disagreement measures rather than treating any single proxy as the object of study.

Our pricing model is related to, but distinct from, that of Goulding et al. (2025), which builds on the frictionless skewness-demand-convexity mechanism of Goulding et al. (2023b) to explore pricing effects of skewness-disagreement interactions, taking disagreement as given by analyst forecast dispersion. Our paper inverts the order of analysis to recover latent disagreement from the joint equilibrium behavior of model variables. This approach yields an interlocking set of distinct model-based predictions about latent disagreement that are not available in single-proxy settings: its nonlinear structure of major proxies, why these proxies disagree, and its signature of the skewness-demand-convexity mechanism itself.

This inversion unlocks analyses in the empirical program of Section 5, including our regime-stability diagnostic, IVOL puzzle account, and two-channel separation. Prior empirical work has documented that the disagreement–return relationship for various proxies is stronger in the presence of short-selling constraints (e.g., Boehme et al., 2006; Nagel, 2005), consistent with the friction-based channel, but has not isolated a friction-independent channel in the same data. Our analyses in Section 5.5 provide new evidence that both channels coexist under the same specification and same sample, and are complementary rather than competing. Their empirical separability is a contribution that is distinct from the theoretical contribution of either channel alone.

Our equilibrium framework also distinguishes the composite from atheoretic aggregation approaches. Huang et al. (2021) construct an aggregate disagreement index via linear partial least squares to predict aggregate market returns. Our approach provides instead a model-based nonlinear composite applicable to individual stocks, with a specific nonlinear structure whose necessity is empirically supported. Bali et al. (2026) represent heterogeneous investors as different machine-learning specifications, each trained on a randomized subsample of over 150 firm characteristics, and measure disagreement as the dispersion across their return forecasts on a broad sample, whereas our measure is derived from a parsimonious model with a small set of theory-motivated inputs, generating testable predictions about regime-stability, the role of frictions, and which proxies are expected to be closely linked to latent disagreement. Internet Appendix E summarizes major categories of disagreement proxies employed in the literature across accounting, economics, finance, and management; see Goulding et al. (2026) for additional discussion and background on each category.

Organization. Section 2 presents the pricing model. Section 3 derives the composite disagreement measure. Section 4 describes the data and empirical construction. Section 5 presents empirical results: portfolio sorts, Fama–MacBeth regressions, the IVOL absorption analysis, the uncertainty-regime analysis, the role of short-selling frictions, the nonlinearity tests, and the characteristics associated with disagreement that DIS detects but that goes undetected by extant proxies. Section 6 offers some conclusions. Appendix A provides proofs, Appendix B develops the multi-asset framework, and Appendix C provides empirical variable details. The Internet Appendix provides a summary of disagreement proxy categories in the literature, additional derivations for equilibrium demand quantities and the multi-asset separation result, higher-order sensitivity results, further robustness analyses, and details on additional empirical variables used in those analyses.

2. Model

We analyze a single-asset framework, which might seem counterintuitive given the framework makes cross-sectional predictions. However, given the nature of idiosyncratic disagreement, we can obtain identical inferences from a single asset framework that separates out of a multi-asset framework with a common factor structure, mutually independent idiosyncratic components, and disagreement about idiosyncratic expected payoffs, in which pricing is governed by the asset's own idiosyncratic statistics (Appendix B). The model of this section is that separated problem and has the benefit of expositional clarity. The parameters that appear below (payout mean, variance, and skewness, and investor beliefs) are all to be understood as idiosyncratic quantities relative to the factor structure, as Appendix B makes explicit. A reader

wishing to start from the multi-asset foundation before proceeding may read Appendix B first; the main text takes the separation result as given and focuses on characterizing the equilibrium relationships among idiosyncratic variables that form the basis of our empirical analysis.

Consider a two-period market having dates $t \in \{0,1\}$ with two assets: a risk-free asset with its price and payoff normalized to one; and one risky asset with price p at date $t = 0$ and payoff $\widetilde{y}$ at date $t = 1$. We follow a standard market-clearing approach to characterize the equilibrium price p. Specifically, payoff $\widetilde{y}$ is exogenously given and the equilibrium price p will be endogenously determined by market clearing of the demands of a continuum of expected-utility maximizing investors, whose utility functions and differences of opinion about $\widetilde{y}$ are also exogenously given.

2.1. Payoff distribution

We model the payoff $\widetilde{y}$ as a random variable with idiosyncratic statistics of mean, variance, and skewness denoted by $\mu = \mathrm{E}[\widetilde{y}]$, $\sigma^2 = \mathrm{Var}[\widetilde{y}]$, and $s = \mathrm{Skew}[\widetilde{y}] = \frac{\mathrm{E}[(\widetilde{y}-\mu)^3]}{\sigma^3}$, respectively, and with marginal cumulative distribution function (cdf) denoted by $F(y) = \mathrm{P}[\widetilde{y} \le y]$. We denote the corresponding net return random variable by $\widetilde{r}$, where $\widetilde{r} = \frac{\widetilde{y}-p}{p}$, with its variance denoted by $\sigma_r^2 = \mathrm{Var}[\widetilde{r}]$.[3] Note that the return mean satisfies $\mathrm{E}[\widetilde{r}] = \frac{\mu-p}{p}$, the return variance satisfies $\sigma_r^2 = \frac{\sigma^2}{p^2}$, and the return skewness is the same as the payoff skewness. We assume μ is finite and σ is strictly positive to avoid a trivial solution.

2.2. Investor utility

There is a continuum of investors, indexed by the unit interval $[0,1]$, each with common utility function of wealth $u(w)$ and initial wealth $w_0 > 0$. Investors prefer more to less, $u'(w) > 0$; are strictly risk averse, $u''(w) < 0$; and have demand for precautionary savings or positive intensity of prudence, $u'''(w) > 0$. Let $\mathcal{A} = -\frac{u''(w_0)}{u'(w_0)}$ denote the coefficient of absolute risk aversion at initial wealth (Pratt, 1964, and Arrow, 1965). Let $\mathcal{P} = -\frac{u'''(w_0)}{u''(w_0)}$ denote the coefficient of absolute prudence at initial wealth (Kimball, 1990).[4] Our assumptions on utility embed the natural conditions of $\mathcal{A} > 0$ and $\mathcal{P} > 0$. This broad class of utility functions includes two of the most widely used classes of utility functions in the literature, namely, constant absolute risk aversion (CARA) utility,

$$u(w) = -e^{-aw}, \qquad a > 0, \tag{1}$$

where a is a scalar that governs absolute risk aversion, and constant relative risk aversion (CRRA) utility,

$$u(w) = \begin{cases} \frac{w^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \ln(w), & \gamma = 1, \end{cases} \tag{2}$$

for $w > 0$, where $\gamma > 0$ is a scalar that governs the degree of relative risk aversion. For CARA, $\mathcal{A} = \mathcal{P} = a > 0$, and for CRRA, $\mathcal{A} = \gamma/w_0 > 0$ and $\mathcal{P} = (1+\gamma)/w_0 > 0$.

2.3. Investor disagreement

Investors have differences of opinion about the expected payoff of the risky asset and, accordingly, also disagree about the expected return. To model their disagreement, we specify separately the relative distribution of beliefs and the scale of those beliefs. This framework encompasses many shapes of belief heterogeneity (polarized, uniform, concentrated-with-tails, or asymmetric) while treating the overall level of disagreement as a separate parameter governed by the scale D.

Specifically, regarding the relative distribution, each investor's belief corresponds to an independent realization of the random variable $\widetilde{\Delta}$ at date $t = 0$, which is a unitless standardized variable governing belief heterogeneity. In particular, $\widetilde{\Delta}$ is a discrete mean-zero random variable with variance normalized to one, which takes $i = 1, 2, \ldots, n$ possible values: $\widetilde{\Delta} = \Delta_i$ with probability π_i. Belief heterogeneity characterizes n types of investors, with proportion π_i of investors being of type i, and models the concentration or dispersion of beliefs relative to each other holding the overall variance fixed to one unit.

Regarding the scale of these beliefs, each type-i investor believes the expected payoff is $\mu_i = \mu + D\Delta_i$, where D is a positive scalar, in units of currency, that expands or contracts the relative distribution of beliefs to their absolute levels. Equivalently, each type-i investor believes that $\widetilde{y}$ is drawn from cdf $F(y - D\Delta_i)$.[5] We refer to D when discussing the role of disagreement in prices or payoffs, which are denominated in units of currency.

Likewise, each type-i investor believes the expected return is $\mathrm{E}[\widetilde{r}] + d\Delta_i$, where $d = \frac{D}{p}$ is a unitless scalar. We refer to d when discussing the role of disagreement in the expected return, which is also unitless. The higher that D (or d) is above zero, the greater the scale of disagreement is within the investor population, all else equal.

2.4. Investor demand and the equilibrium price

Each investor of type $i = 1, 2, \ldots, n$ demands $x_i(p)$ units of the risky asset given price p, where $x_i(p)$ maximizes investor i's expected utility of date–1 wealth,

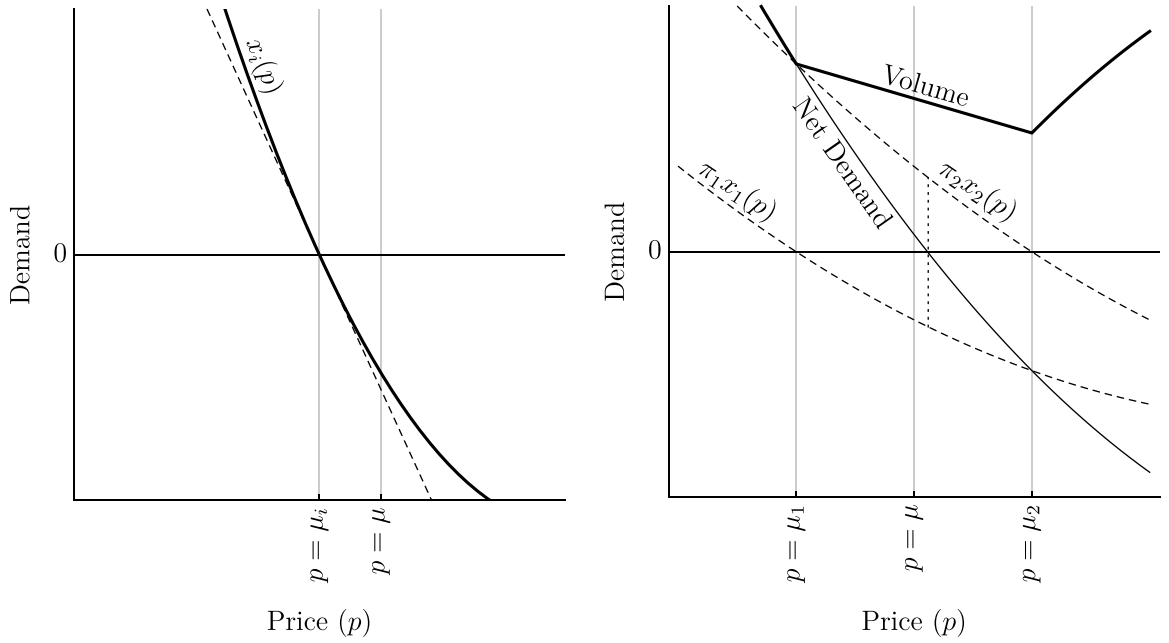
$$x_i(p) = \arg\max_x \mathrm{E}_i[u(w_0 + x(\widetilde{y} - p))], \tag{3}$$

where E_i denotes the expectation is taken with respect to investor i's beliefs: $F(y - D\Delta_i)$. Under type-i investor's belief, the expected payoff is $\mu_i = \mathrm{E}_i[\widetilde{y}] = \mu + D\Delta_i$. For notational simplicity, we denote the variance under this investor's belief by $\sigma_i^2 = \mathrm{E}_i[(\widetilde{y} - \mu_i)^2]$ and the skewness by $s_i = \frac{\mathrm{E}_i[(\widetilde{y}-\mu_i)^3]}{\sigma_i^3}$. It can be shown that variance and skewness do not depend on i because investors only disagree about the mean payoff via a shift of the cdf, $F(y - D\Delta_i)$. As a result, we drop the subscripts and write $\sigma = \sigma_i = \mathrm{SD}[\widetilde{y}]$ and $s = s_i = \mathrm{Skew}[\widetilde{y}]$.

[3] Division by p is well-defined, for example, under the natural condition of limited liability that entails the payoff $\widetilde{y}$ has only nonnegative support such that $p > 0$ in equilibrium.

[4] Leland (1968) and Sandmo (1970) connect a positive third derivative of a utility function to a preference for precautionary savings and Kimball (1990) indicates the coefficient of absolute prudence or intensity of prudence measures the strength of the precautionary saving motive.

[5] For example, suppose the expected payoff is $\mu = 10$. Consider the case of $n = 2$ with $\Delta_1 = -1$ and $\Delta_2 = 1$ equally likely. These Δ_i's are zero on average with unit variance, representing that type-1 investors are relatively "pessimistic" and type-2 investors are relatively "optimistic". If the scale of disagreement is $D = 0.5$, then each pessimistic investor believes the payoff distribution of the risky asset is shifted by -0.5 such that $\mu_1 = 9.5$ and therefore that $\widetilde{y}$ is drawn from cdf $\mathrm{P}[\widetilde{y} - 0.5 \le y] = \mathrm{P}[\widetilde{y} \le y + 0.5] = F(y + 0.5)$. Similarly, each optimistic investor believes the payoff distribution of the risky asset is shifted upward by 0.5 such that $\mu_2 = 10.5$ and therefore that $\widetilde{y}$ is drawn from cdf $\mathrm{P}[\widetilde{y} + 0.5 \le y] = \mathrm{P}[\widetilde{y} \le y - 0.5] = F(y - 0.5)$.



(a) Investor type i demand, pos. skew. (b) Net demand and trading volume, pos. skew.

Fig. 1. Investor demand and trading volume as a function of price.
Notes: Figure (a) plots the second-order Taylor polynomial of the demand $x_i(p)$ of a type-i investor with belief $\mu_i = \mu + D\Delta_i$ as a function of price (solid curve) and its tangent line at $p = \mu_i$ (dashed line) over the domain $p \in [\mu_i - \sigma, \mu_i + \sigma]$, where $\Delta_i = -1$, under CRRA utility with relative risk aversion parameter γ. Figure (b) plots the trading volume (bold solid curve), the second-order Taylor polynomials of the total demand functions of type-1 and type-2 investors with beliefs $\mu_1 = \mu + D\Delta_1$ and $\mu_2 = \mu + D\Delta_2$ (dashed curves), respectively, as well as the net demand (solid curve) over the domain $p \in [\mu - 0.5\sigma, \mu + 0.5\sigma]$, where $\Delta_1 = -1$, $\Delta_2 = 1$, $\pi_1 = \pi_2 = 0.5$. The vertical dotted line to the right of μ marks the equilibrium price at which type-1 and type-2 demands are equidistant from the price axis, so that positive and negative demands offset to clear the market. The figures have the following parameter values in common: $D = 0.05$, $\mu = 1$, $\sigma = 0.2$, $s = 0.6$, $\gamma = 2$, $w_0 = 1$.

Proposition 1 (*Demand*). *The Taylor expansion of type-i's demand for the risky asset around $p = \mu_i$ is*

$$x_i(p) = -\frac{1}{\mathcal{A}}\frac{1}{\sigma^2}(p - \mu_i) + \frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}^2}\frac{s}{\sigma^3}(p - \mu_i)^2 + h_i(p)(p - \mu_i)^2, \tag{4}$$

where $h_i(p)$ is an unknown function that converges to 0 as $p \to \mu_i$ by Taylor's theorem.[6]

Proposition 1 characterizes type-i demand via a Taylor expansion around $p = \mu_i$. Demand is zero at $p = \mu_i$, decreasing in price with slope $-\frac{1}{\mathcal{A}\sigma^2}$, and shaped by skewness through the quadratic term, whose influence is stronger the higher $\mathcal{P}$ and weaker the higher $\mathcal{A}$.

2.4.1. Skewness and local demand convexity

Fig. 1(a) illustrates under CRRA utility the main properties of the demand function in the case of positive skewness: $s > 0$. Positive skewness produces nonlinear demand and, in particular, locally strictly convex demand around the investor's belief (solid curve in Fig. 1(a)). This convexity generates asymmetric demand responses (to symmetric deviations in price) for all investor types. For example, for prices below the investor's belief, $p < \mu_i$, the investor has increasingly positive demand that accelerates as price decreases; whereas, for prices above, $p > \mu_i$, the investor has negative demand or short interest in the risky asset, which increases in magnitude at a diminishing rate as price increases because of the convexity.[7] Without skewness, demand would be a linear function (dashed tangent line in Fig. 1(a)), which is the norm in the classic set-up of a normally-distributed payoff and CARA utility, and both positive demand and short interest would increase in magnitude symmetrically away from $p = \mu_i$ at a uniform rate.

2.4.2. Market-clearing equilibrium price and expected return

Combining the demand functions of all investor types in a market-clearing analysis determines the equilibrium price. Because investors of type i each trade $x_i(p)$ and are of size π_i in the population, we can express the net demand of all investors as

$$\pi_1 x_1(p) + \pi_2 x_2(p) + \cdots + \pi_n x_n(p), \tag{5}$$

where $\pi_i > 0$ and $\sum_{i=1}^{n} \pi_i = 1$. We assume that the net supply of the risky asset is $X = 0$ and determine the equilibrium price from market clearing: $\sum_{i=1}^{n} \pi_i x_i(p) = X$. Note that we do not require noisy supply to motivate trade as in market models with investors having common expectations. Differences in beliefs are sufficient to generate trade.

Fig. 1(b) illustrates under CRRA utility the net demand (5) in a market with two types of investors who disagree: $\pi_1 = \frac{1}{2}$ of investors are type-1 investors, who are relatively pessimistic about the expected payoff, $\mathrm{E}_1[\tilde{y}] = \mu - D$, and $\pi_2 = \frac{1}{2}$ of investors are type-2 investors, who are relatively optimistic about the expected payoff, $\mathrm{E}_2[\tilde{y}] = \mu + D$.

[6] A second-order Taylor expansion is sufficient to capture first- and second-order price sensitivity as well as generate the property of local convexity of demand generated by payout skewness. Higher-order expansions that reflect additional moments, such as kurtosis, could reduce approximation error for prices outside the range of investor beliefs, but at the expense of increased complexity. To preview, the equilibrium price of *Proposition* 2 falls within the span of beliefs for disagreement not too large, and expected returns have limited sensitivity to fourth or higher moments empirically. The composite measure DIS inherits this approximation; its empirical performance across subsamples (Section 5 and Internet Appendix I) provides indirect evidence that the second-order expansion captures the economically relevant variation.

[7] This demand behavior reflects the preference for precautionary savings, for example, represented by CRRA utility of a positively skewed payoff.

Fig. 1(b) plots the case of a positively skewed payoff ($s = 0.6$). The total demands of each investor type, $\pi_1 x_1(p)$ and $\pi_2 x_2(p)$, respectively, are decreasing and convex in price p in a neighborhood overlapping their beliefs, by Proposition 1. In Fig. 1(b), this overlapping region of prices includes $p = \mu$, which is between $p = \mu_1$ and $p = \mu_2$, and a price above $p = \mu$ that perfectly offsets demands to clear the market. Again, because investors are risk averse, they are unwilling to trade a deterministic amount p for an uncertain payoff that equals p in expectation. Hence, demand is zero for all type-1 investors at price $p = \mu_1$. At $p = \mu_1$, type-2 investors, who believe the asset is worth more than p in expectation, demand a positive quantity, and therefore, net demand is positive. As price increases above $p = \mu_1$, type-1 demand becomes negative to offset the positive demand of type-2, increasing short interest and reducing net demand.

More generally, setting net demand (5) equal to zero determines the equilibrium price that offsets demands to clear the market. Because each $x_i(p)$ depends on the scale of disagreement, D, through belief $\mu_i = \mu + D\Delta_i$, the equilibrium price also depends on D. In Proposition 2, we use implicit differentiation of price in the market-clearing condition with respect to D to obtain a Taylor expansion in terms of the scale of disagreement, for D not too large.[8]

Proposition 2 (*Equilibrium Price*)**.** *The Taylor expansion of the equilibrium price p as a function of the scale of disagreement, around $D = 0$, is given by:*

$$p(D) = \mu + \frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}D^2 + h(D)D^2, \tag{6}$$

where $h(D)$ is an unknown function that converges to 0 as $D \to 0$ by Taylor's theorem.[9]

Proposition 2 shows how the equilibrium price depends on the scale of disagreement.[10] If $s > 0$, then equilibrium price increases in D, is larger than expected payoff ($p > \mu$), and by the second-order Taylor polynomial in (6), the expected return satisfies

$$\mathrm{E}[\tilde{r}] = -\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma_r}d^2, \tag{7}$$

which is decreasing in d. (Recall $d = D/p$ and $\sigma = \sigma_r p$.) By inspection of (6) and (7), we find that more disagreement means higher prices, and, therefore, lower expected returns for positive skewness, all else equal. The influence of disagreement on expected returns is stronger the higher the demand for precautionary savings $\mathcal{P}$ and weaker the higher the risk aversion $\mathcal{A}$. Intuitively, when $\mathcal{A}$ is higher, investors are more reluctant to take larger risky positions, reducing how aggressively buyers and sellers act on their views: disagreement translates into smaller price impact. When $\mathcal{P}$ is higher, investors react more sensitively to uncertainty by shifting toward safer positions, which moves prices more, and expected returns adjust more in equilibrium. The model identifies the $\mathcal{P}/\mathcal{A}$ ratio, the ratio of absolute prudence to absolute risk aversion, as the coefficient governing how strongly disagreement depresses expected returns. Standard models either use CARA utility, where $\mathcal{P} = \mathcal{A}$ and the ratio is identically one, or ignore higher-order preferences entirely. Our framework makes the two separable. This generalization matters theoretically because it identifies a structural parameter: the disagreement–return relationship is governed by the interaction of prudence and risk aversion, not by risk aversion alone. In our empirical application, however, the $\mathcal{P}/\mathcal{A}$ ratio does not drive the cross-sectional predictions we test, since $\mathcal{P}/\mathcal{A}$ enters as a scalar multiplier common to all assets.

Note that because of the standard simplifying assumption of zero supply, the expected return (7) is zero in the case of zero skewness. This result reflects the fact that positive expected returns typically are the result of a risk premium necessary to induce investors to have net positive demand for the asset in equilibrium. Therefore, with zero supply, we have isolated the impact of disagreement on expected returns net of the risk premium rather than gross of the risk premium. Our empirical tests, which examine cross-sectional variation in returns conditional on standard factor controls, are designed to isolate precisely this deviation, making the zero-supply normalization appropriate for the cross-sectional predictions we test.

2.4.3. Demand convexity and the disagreement–price relationship

The local demand convexity established in Section 2.4.1 has a direct implication for equilibrium pricing. Because the distribution of investor beliefs is centered around fundamental value, a price at fundamental value cannot offset the asymmetric demand responses documented above: positive demand exceeds short interest in magnitude at any given deviation from the investor's belief. The market-clearing price must therefore be pushed higher. For example, in Fig. 1(b), net demand remains positive at $p = \mu$; price must increase above μ to clear the market. Higher scales of disagreement widen investor beliefs, amplify the asymmetry, and push the market-clearing price further above fundamental value.

Without demand convexity, as would obtain with zero skewness, long and short positions would perfectly offset at fundamental value regardless of the scale of disagreement. The departures from this benchmark documented in Proposition 2 are therefore entirely attributable to the interaction of skewness and disagreement.

This self-insulating property of disagreement-driven overpricing, in which the same skewness preferences that generate mispricing also discourage the arbitrage trades that would correct it, provides a preference-based complement to friction-based explanations for the persistence of anomaly returns (Shleifer and Vishny, 1997; Pontiff, 2006). The distinction is structural. In friction-based limits to arbitrage, rational agents are prevented from correcting mispricing by exogenous costs. In our framework, rational unconstrained agents endogenously choose smaller corrective positions because the short side of a positively skewed asset is itself negatively skewed.

2.5. Trading volume

Denote trading volume as a function of price p by $V(p) = \int_0^1 |x_i(p)|di$; i.e., the sum of the absolute value of investors' trades. Because investors of type i each trade $x_i(p)$ and are of size π_i in the population, we have trading volume

$$V(p) = \pi_1|x_1(p)| + \pi_2|x_2(p)| + \cdots + \pi_n|x_n(p)|, \tag{8}$$

where $\pi_i > 0$ and $\sum_{i=1}^n \pi_i = 1$. Fig. 1(b) shows trading volume as a function of price for a market with two investor types. There are two kinks in the trading volume curve: one occurs where type-1 investors' demand crosses zero (at μ_1) and the other occurs where type-2 investors' demand crosses zero (at μ_2).

We now write x_i and V to denote the equilibrium demand of investor type i and equilibrium volume, respectively, in which each of $x_i(p)$ and $V(p)$ are evaluated at equilibrium price $p(D)$, using the second-order Taylor polynomials for demand, volume, and price:

$$x_i = \frac{1}{\mathcal{A}}\frac{1}{\sigma^2}D\left(\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}D - \Delta_i\right)\left[-1 + \frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}D\left(\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}D - \Delta_i\right)\right], \tag{9}$$

[8] The phrase "for D not too large" means that there exists a $\overline{D} > 0$ such that the approximation is reasonable for all $D < \overline{D}$. See Internet Appendix F for additional discussion in the context of equilibrium volume and price relationships.

[9] The second-order Taylor polynomial approximation of equilibrium price captures all disagreement sensitivity inherited from the second-order demand polynomials of *Proposition* 1 because of its determination via market clearing of such demands. That is, higher-order expansions for equilibrium price are relevant only for higher-order expansions of demand.

[10] Note that the asset's idiosyncratic statistics are relevant for pricing rather than, say, covariances with the market or aggregate-level disagreement, consistent with the interpretation given in Appendix B.

$$V = \pi_1|x_1| + \pi_2|x_2| + \cdots + \pi_n|x_n|, \tag{10}$$

where $\pi_i > 0$ and $\sum_{i=1}^{n} \pi_i = 1$.

Relatedly, short interest is the absolute value of all negative demands. In the model, short interest is determined in equilibrium as a function of all other parameters and variables. It ends up being in constant proportion ($\frac{1}{2}$) to equilibrium volume because of our simplifying assumption of zero net demand, but it varies according to model parameters. Specifically, $\sum_{x_i<0} \pi_i|x_i| = -\sum_{x_i<0} \pi_i x_i = \sum_{x_i>0} \pi_i|x_i|$, so short interest is volume divided by two, and therefore, volume and short interest will have common sensitivities to other variables in the model. In our model, it is negative demand, or short interest, that translates the convexity of demand for a positively-skewed asset into an equilibrium price above fundamental value. So, although our analysis focuses on overall volume for simplicity of exposition, the sensitivity of volume in the model is interchangeable with that of short interest. Accordingly, we use a measure of short interest as our volume proxy in our empirical implementation.

Note that x_i in (9) depends on disagreement both via the underlying source of heterogeneity, Δ_i, and the overall scale of disagreement, D. Likewise, V depends on the scale of disagreement. To understand the equilibrium relationship between volume and the scale of disagreement, consider Fig. 1(b). At the market-clearing price, denoted by the vertical dotted line, trading volume is decreasing in price; i.e., an increase in price results in lower overall trading activity. If an increase in equilibrium price is the result of a higher scale of disagreement, however, demands expand away from the price axis at a rate faster than the trading volume decreases in price. The result is that equilibrium trading volume (as well as short interest) increases in the scale of disagreement. This result is consistent with other models in the literature, but we arrive at this result based on investors with different beliefs trading a skewed asset in a frictionless model, which to our knowledge is a novel mechanism. See Proposition 3 and Internet Appendix F for a rigorous analysis.

Proposition 3. *Equilibrium trading volume, V, is increasing in the scale of disagreement, D:* $\frac{\partial V}{\partial D} > 0$.

Proposition 3 indicates that trading volume (and short interest) is a proxy for disagreement, consistent with the implications of earlier literature and the use of volume-based measures as proxies for disagreement, but arrived at under different assumptions and, hence, provides new support for the volume-based category of proxies. Higher levels of disagreement are associated with higher levels of trading volume and short interest, all else equal. See Internet Appendix F for more discussion of the intuition for this result.

3. A composite proxy of disagreement

In this section, we apply the model of Section 2 to develop a novel composite proxy of investor disagreement as a function of all model variables plus variables related to analyst forecast dispersion.

3.1. Two symmetric investor types

First, we analyze the special case of disagreement between $n = 2$ symmetric investor types. Let $\pi_1 = \pi_2 = \frac{1}{2}$, $\Delta_1 = -1$, and $\Delta_2 = 1$, so that $\mathrm{E}[\widetilde{\Delta}] = 0$ and $\mathrm{Var}[\widetilde{\Delta}] = 1$. Because of market clearing, we have $x_1 < 0$ and $x_2 > 0$, and so $V = -\frac{1}{2}x_1 + \frac{1}{2}x_2$. Substituting (9) for x_1 and x_2 and simplifying, we find that trading volume equals

$$V = \frac{1}{\mathcal{A}}\frac{1}{\sigma^2} D\left[1 - \frac{1}{2}\frac{\mathcal{P}^2}{\mathcal{A}^2}\frac{s^2}{\sigma^2}D^2\right]. \tag{11}$$

Eq. (11) relates all key model variables in a simple expression. This expression relates volume to disagreement, volatility, and skewness, but does not yet incorporate the analyst forecast channel that our model identifies as a distinct amplifier of disagreement. We turn to this next.

3.2. Analyst dispersion

We now augment our main model with a simple auxiliary model that incorporates the impact of analyst forecasts on the scale of disagreement. Analyst forecast dispersion and investor disagreement are correlated but distinct. Recall that investors agree about the payoff variance (σ^2) and, therefore, any impact of analyst forecast information on investor beliefs manifests in the variation of their means. Also, recall that the distribution of expected payoffs corresponds with $\mu + D\widetilde{\Delta}$, where $\widetilde{\Delta}$ is a standardized random variable that reflects relative differences and D governs scale. Analyst forecast dispersion thus impacts the scale of disagreement via a public information effect that does not alter the common payoff volatility belief, σ, and does not necessarily alter the underlying relative investor heterogeneity captured by $\widetilde{\Delta}$. Accordingly, we model the variance of means indexed to the common volatility level, $(\mathrm{Var}[(\mu + D\widetilde{\Delta})/\sigma])$, as proportional to the information conveyed by analyst forecasts via the posterior variance of the mean payoff given analyst forecasts. With this approach, analyst forecast dispersion will merge conveniently with our main model.

To fix ideas, let $j = 1, 2, \ldots, m$ denote financial analysts who each obtain a noisy signal about the true expected payoff and publish a corresponding forecast. These forecasts are observed by investors at date $t = 0$. Let $\widetilde{\mu} \sim \mathcal{N}(\mu, \sigma_\mu^2)$ be a random variable denoting the unknown expected payoff of the risky asset; i.e., $\widetilde{\mu}$ is normally distributed around the payoff's true mean μ with variance σ_μ^2. We refer to σ_μ as fundamental uncertainty: the dispersion of beliefs about the true mean payoff that is present independent of analyst forecasts. Via their research process, analysts obtain noisy observations of $\widetilde{\mu}$. Suppose analyst j publishes a forecast A_j, which is a realization of the random variable $\widetilde{A}_j = \widetilde{\mu} + \widetilde{a}_j$, where $\widetilde{a}_j \sim \mathcal{N}(0, \sigma_a^2)$ is independent, normally distributed noise with variance σ_a^2. Hence, analyst forecasts are correlated with each other because their signals partially reflect the true payoff $\widetilde{\mu}$, but each forecast differs because of analyst-specific noise; i.e., $\widetilde{a}_j$. Analysts publish forecasts conveying their noisy inferences about μ and the precision of their information is governed by $\frac{1}{\sigma_a^2}$.

Lemma 4. *Suppose the variance of investor beliefs about the expected payoff per unit of volatility is proportional to the variance of the expected payoff conditional on analysts' forecasts:* $\mathrm{Var}[(\mu + D\widetilde{\Delta})/\sigma] \propto \mathrm{Var}\left[\widetilde{\mu}\middle|\widetilde{A}_1, \widetilde{A}_2, \ldots, \widetilde{A}_m\right]$. *Then,*

$$\frac{D^2}{\sigma^2} = \left(\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}\right)^{-1}, \tag{12}$$

when the proportionality constant is one.

Under the simplifying assumptions of analysts' signals, the scale of disagreement depends on the number of analysts, m, in Lemma 4. Because we model each analyst's signal as having independent noise for simplicity, analysts' collective forecasts reduce the posterior variance of the true payoff. If instead the noise in analyst signals were positively correlated across analysts or if the number of analysts increased with the uncertainty around the mean payoff, which are both plausible alternative assumptions, then the role of the number of analysts would drop out or be diminished. In our empirical implementation, we scale the number of analysts each month by the cross-sectional average number of analysts to allow for some variation across firms but to moderate the impact coming from our simplifying assumptions as well as to keep our disagreement measure unitless (see Section 4).

By Lemma 4, in the absence of the influence of analyst forecasts ($m = 0$), risk-adjusted disagreement is captured by the fundamental uncertainty about μ: $D/\sigma = \sigma_\mu$. Thus, total heterogeneity among investors has a source that does not depend solely on heterogeneity of analysts (via σ_a^2) and instead reflects both idiosyncratic volatility (i.e., σ) and fundamental uncertainty about the mean (i.e., σ_μ). When there are two or more analysts, however, dispersion in analysts' forecasts can serve

to amplify the magnitude of already-existing differences in investors' beliefs.[11] Specifically, based on the method-of-moments estimate of the variance of analyst forecasts, $\widehat{\sigma}_a^2$, the expression (12) characterizes explicitly the relationship between the scale of disagreement among investors, D, and analyst forecast dispersion, $\widehat{\sigma}_a$, as measured by the standard deviation of analyst forecasts. Although in this simple model, D and $\widehat{\sigma}_a^2$ are functionally related, analyst dispersion in the data is an imperfect proxy for disagreement. We can combine this result with our main model to capture interactions with trading-related components.

3.3. Composite disagreement measure

We now merge this simple auxiliary analyst dispersion model with our main trading model to develop a composite proxy of disagreement. Substituting Eq. (12) for D^2/σ^2 into the second term of (11) yields

$$V = \frac{1}{\mathcal{A}}\frac{1}{\sigma^2}D\left[1 - \frac{1}{2}\frac{\mathcal{P}^2}{\mathcal{A}^2}s^2\left(\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}\right)^{-1}\right]. \tag{13}$$

In our stylized model, supply and, therefore, shares outstanding are zero. In empirical applications of our model, however, shares outstanding are positive for each firm. To aid empirical implementation, let v denote volume traded as a fraction of shares outstanding (which is unitless) and suppose that aggregate initial wealth of all investors, w_0, is proportional to the market value of the risky asset, $w_0 = pX$, with unit proportionality constant. Applying identities $\sigma_r p = \sigma$ and $d = D/p$, normalizing initial wealth to one, and rearranging (13) to isolate d yields

$$d = \mathcal{A}\sigma_r^2 v\left[1 - \frac{1}{2}\frac{\mathcal{P}^2}{\mathcal{A}^2}s^2\left(\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}\right)^{-1}\right]^{-1}, \tag{14}$$

which is a single equation relating our unitless disagreement variable to all quantities of interest. Define $g \equiv \frac{1}{2}\frac{\mathcal{P}^2}{\mathcal{A}^2}s^2\left(\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}\right)^{-1}$. Note that the term $\left[1-g\right]^{-1}$ in (14) should be positive in order for d to be well-defined. This condition will be met as long as the term g is between 0 and 1. Note that $g \geq 0$ because the number of analysts is nonnegative ($m \geq 0$), the variance of analyst forecasts is nonnegative ($\sigma_a^2 \geq 0$) and the variance of the true payoff is positive ($\sigma_\mu^2 > 0$). However, depending on the scale of the quantities $\mathcal{A}$, $\mathcal{P}$, s, m, σ_a, and σ_μ, then g can potentially exceed 1. To ensure the condition $g \in [0,1]$, we apply a standard monotonic transformation $g \to 1 - e^{-g}$, which is increasing in g, equal to 0 for $g = 0$, and equal to 1 in the limit as $g \to \infty$. This preserves the essential multiplicative structure while keeping $g \in [0,1]$ and is robust for empirical implementation. Applying this to (14), after simplification we obtain the modified composite disagreement score:

$$d^* = \mathcal{A}\sigma_r^2 v \cdot \exp\left[\frac{\frac{1}{2}\frac{\mathcal{P}^2}{\mathcal{A}^2}s^2}{\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}}\right]. \tag{15}$$

We will use (15) to construct a unitless disagreement proxy for our main empirical analyses. Specifically, we use DIS $= \ln(1 + d^*)$, where d^* is as in (15), which helps to attenuate the skewness of d^* exhibited in our empirical implementation.

Economic interpretation. The composite measure of (15) is the product of a volatility-volume term, $\mathcal{A}\sigma_r^2 v$, and an exponential amplification term governed by skewness and analyst forecast dispersion. This multiplicative structure means that d^* registers high disagreement only when multiple components are simultaneously elevated: high idiosyncratic volatility alone, or high volume (short interest) alone, does not produce a high score unless the other components corroborate. For example, if σ_r rises broadly across stocks, as it does in periods of aggregate stress, but v and the dispersion-related amplification term do not rise commensurately, then d^* remains moderate. An additive composite, by contrast, could register elevated disagreement whenever any single component is high, regardless of the others. The co-occurrence requirement embedded in the multiplicative form thus provides a built-in filter against variation in any one component that is unrelated to firm-specific disagreement. Whether this structural property translates into empirical advantages over linear alternatives is a testable question, which we examine in Section 5.

3.4. Sensitivities

The composite proxy (15) is increasing in relative risk aversion, idiosyncratic volatility, idiosyncratic skewness, trading volume, analyst forecast dispersion, and in fundamental uncertainty about the mean payoff, all else equal; and is decreasing in the precision of analysts' information and the number of analysts, all else equal. Moreover, this composite proxy reflects model interactions among these variables, which are missing from individual measures and otherwise difficult to account for in empirical analyses.

Disagreement d^* in (15) is a function of $\mathcal{A}$ (coefficient of absolute risk aversion), $\mathcal{P}$ (coefficient of absolute prudence), σ_r (idiosyncratic return volatility), v (short interest), s (idiosyncratic return skewness), m (number of analyst forecasts), σ_a (dispersion of analyst forecasts), and σ_μ (fundamental uncertainty) with the following sensitivities to key variables with observable proxies:

$$\frac{\partial d^*}{\partial \sigma_r} = 2\frac{d^*}{\sigma_r} > 0, \tag{16}$$

$$\frac{\partial d^*}{\partial v} = \frac{d^*}{v} > 0, \tag{17}$$

$$\frac{\partial d^*}{\partial \sigma_a} = d^*\frac{g}{\left(\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}\right)}\frac{2m}{\sigma_a^3} > 0. \tag{18}$$

Hence, idiosyncratic volatility, trading volume, and analyst forecast dispersion are each proxies for disagreement, all else held equal. Moreover, each of the sensitivities (16) and (17) is consistent with those implied by (11) of the main model. In Internet Appendix H, we explore higher-order sensitivities.

An important feature of (15) is that the identity and role of each component, the specific nonlinear interactions among them, and the absence of any additional variables are all determined by the conditions of the model. This distinguishes its empirical counterpart, DIS, from empirical composites whose components and weights are chosen to optimize in-sample performance.

4. Data and construction of composite disagreement DIS

Our primary dataset is the intersection of CRSP and COMPUSTAT for all common stocks traded on the NYSE, AMEX, or NASDAQ. Following standard practices, we exclude stocks with month-end prices below $5, require at least 15 valid daily observations per stock-month, and exclude stock-months with 10 or more zero daily returns. Because our composite measure incorporates analyst forecast dispersion, we restrict the sample to stocks covered by at least two analysts, using data from the I/B/E/S unadjusted details file. Analyst forecast dates are less reliable before 1994 (Clement and Tse, 2005; Kirk et al., 2014) and 2022 is the last year of available data for analyst forecasts and for our expected idiosyncratic skewness measure.[12] The resulting sample contains 719,940 monthly observations spanning January 1994 to December 2022.

[11] Theoretically, this amplification is true for $m = 1$, but empirically we need $m \geq 2$ to estimate σ_a^2.

[12] The ISKEW measure is obtained from the authors of Boyer et al. (2010), who extended it through 2022.

4.1. Composite disagreement DIS

We construct DIS following the form of Eq. (15) under CRRA utility with relative risk aversion parameter γ.[13] Each component is described below and all variables are constructed using only information available at the time of portfolio formation; see Appendix C for complete variable definitions.

Risk aversion. We set $\gamma = 2$, consistent with common values in the literature. Because $\mathcal{P}/\mathcal{A}$ enters (15) under CRRA utility only through the scalar multiplier $(\mathcal{P}/\mathcal{A})^2 = (1 + 1/\gamma)^2$, it scales the exponential amplification term uniformly across all stocks. Tables I.5 and I.6 in the Internet Appendix present evidence consistent with the interaction structure of the components, not the preference calibration, driving cross-sectional predictions: DIS-decile spread CAPM alphas are stable and Fama–MacBeth coefficients are uniformly significant at the 1% level across $\gamma \in \{0.5, 1, 2, 4, 8\}$.

Idiosyncratic volatility. We compute IVOL as the standard deviation of daily residuals from a Fama and French (1993) three-factor model, following Ang et al. (2006), expressed in percentage: $\sigma_r = \text{IVOL}_{i,t}$.

Trading volume. Volume is measured as monthly short interest, SIR, from the COMPUSTAT short interest file, the ratio of shares shorted to shares outstanding (Chen et al., 2019; Karpoff and Lou, 2010), adding one to ensure strict positivity. We supplement with NASDAQ short interest data prior to 2003 (Chen et al., 2019, 2022). Hence $v = 1+\text{SIR}_{i,t}$. Section 2.5 motivates SIR as the volume proxy rather than turnover because the model predicts that short interest and volume move proportionally in equilibrium, and SIR more directly reflects the pessimistic demand that generates the overpricing mechanism. Section 5.8 (Table I.9 in the Internet Appendix) shows that turnover-based measures (TO, DTO, SUV) are also positively associated with DIS.

Expected idiosyncratic skewness. Because realized skewness lacks persistence (Harvey and Siddique, 1999, 2000), we use the expected idiosyncratic skewness metric ISKEW of Boyer et al. (2010), derived from 60-month firm characteristics. Hence $s = \text{ISKEW}_{i,t}$.

Analyst forecast dispersion. We use the Diether et al. (2002) measure: the standard deviation of fiscal-year-end analyst EPS forecasts scaled by the mean absolute EPS forecast, expressed in percentage.[14] Hence $\sigma_a = \text{AFD}_{i,t}$. While Cen et al. (2017) show that the choice of normalization scalar can inject return-predictive information into the level dispersion measure, AFD enters our composite as an amplifier rather than a standalone predictor, so our construction is not exposed to this concern.

Number of analysts. ANALYSTADJ is the number of analysts with valid forecasts for stock i in month t divided by the cross-sectional average number of analysts per firm in the same month, computed over firms with at least two analysts. This normalization also makes the measure unitless and comparable across time periods with different analyst coverage norms.

Return volatility. We proxy for fundamental uncertainty using one-month ex-post return volatility based on daily data, expressed in percentage: $\sigma_\mu = \text{RVOL}_{i,t}$.[15] Although σ_r (IVOL) and σ_μ (RVOL) are both return-volatility-based, they enter (15) through different channels: σ_r scales the multiplicative volatility–volume prefactor $\mathcal{A}\sigma_r^2 v$, whereas σ_μ enters through the denominator $\frac{m}{\sigma_a^2} + \frac{1}{\sigma_\mu^2}$ of the exponential amplifier, governing how strongly analyst forecast dispersion amplifies existing belief heterogeneity.

Composite disagreement DIS. Our composite measure is $\text{DIS} = \ln(1 + d^*)$, where d^* is as defined in Eq. (15), specialized to CRRA utility with the above empirical proxies. The logarithmic transformation retains nonnegativity and attenuates the right-skewness of d^*.

4.2. Other variables

In addition to measuring disagreement and its impact on subsequent returns, we examine several firm- and stock-specific characteristics that explain the cross-sectional variation in returns which are also used as control variables in later analyses. We highlight the most important ones in this section. LnSIZE is computed as the natural logarithm of market capitalization and BTM as the book-to-market ratio as in Davis et al. (2000). Moreover, to account for the variations in returns as shown in the empirical asset pricing literature, we include MOM (Carhart, 1997) in our analysis defined as the cumulative returns from month $t-12$ to $t-1$. Notably, we omit the lagged return from MOM, in line with Jegadeesh and Titman (1993). BETA is the coefficient derived from regressing a stock's daily returns in month t on the market return. ILLIQ is based on Amihud's (2002) illiquidity measure which averages the ratio of absolute daily returns to daily dollar trading volume. Lastly, to account for the payoff of lottery stocks, we use RMAX, which is the average of the five highest daily returns in a given month t (Bali et al., 2017). All variables, including our composite disagreement measure, are evaluated at a monthly level. A comprehensive overview of all variables utilized in this paper is provided in Appendix C.

4.3. Descriptive statistics

First, Table 1 provides a descriptive overview of our composite disagreement measure, DIS, and selected characteristics of our sample. Note that stocks in our sample are relatively large because of our restriction to two or more analysts, resulting in average market size of \$7.86B, which is more than 60% larger than the average market size for stocks without the analyst restrictions. Focusing on our key variable of interest, DIS, the sample exhibits an average disagreement of 11.99 with an average standard deviation of 34.17. Notably, the distribution of this measure is right-skewed, with the 10th percentile at 1.04 and 90th percentile at 24.08. Most stocks exhibit positive ISKEW and each of the variables in Table 1 is itself positively skewed (i.e., with its mean above its median). In particular, analyst forecast dispersion, AFD, a widely used measure of disagreement, is especially right-skewed.

To further investigate the interplay between our empirical proxies for the components of our disagreement measure, Panel A of Table 2 presents pooled correlations of DIS with its individual components and various stock and firm characteristics, pooling all firm-months. In line with our model's predictions, there is a positive correlation between the measure and each of its individual components. DIS comoves relatively strongly with IVOL (0.47) and relatively weakly with AFD (0.10) and SIR (0.12). To the extent that DIS is an accurate proxy of disagreement, of the major disagreement proxies, IVOL is then the most aligned with disagreement and, therefore, we should expect DIS to absorb some of the return predictability of IVOL in empirical tests. We analyze this connection in Section 5.3. DIS also comoves relatively strongly with ISKEW (0.51), reflecting the important role of skewness in our model.

Panel B of Table 2 presents correlations among the major disagreement proxies, AFD, IVOL, and SIR. These cross-correlations are relatively low, highlighting the disagreement among different disagreement proxies.[16] These near-zero cross-correlations are the empirical face of the disagreement of disagreement: the proxies that purport to measure the same latent variable share almost none of their variation.

[13] Under CRRA utility at unit initial wealth, $\mathcal{A} = \gamma$ and $\mathcal{P} = 1 + \gamma$ so that $\frac{\mathcal{P}^2}{\mathcal{A}^2} = \left(1 + \frac{1}{\gamma}\right)^2$.

[14] Following Hribar and McInnis (2012), we exclude observations with mean absolute forecast below \$0.10 EPS. We exclude stopped and excluded estimates and carry forward estimates released up to 105 days before the earnings announcement.

[15] Alternatives such as profitability-based measures would reduce sample size and introduce sensitivity to the choice of metric. Simple return volatility is widely available and offers a direct reflection of perceived variability.

[16] In Table I.1 of Internet Appendix I, we report time-series average correlations corresponding to those of Table 2, from which we draw similar conclusions.

Table 1
Descriptive statistics.

Variable	Code	Mean	SD	p10	p25	p50	p75	p90
Composite disagreement	DIS	11.99	34.17	1.04	1.79	3.48	8.78	24.08
Short interest	SIR	0.05	0.06	0.00	0.01	0.03	0.06	0.11
Idiosyncratic volatility	IVOL	2.10	1.46	0.81	1.14	1.71	2.62	3.84
Exp. idiosyncratic skewness	ISKEW	0.80	0.58	0.15	0.45	0.76	1.07	1.46
Analyst forecast dispersion	AFD	13.18	64.07	0.74	1.51	3.58	9.61	25.24
Market size ($ billions)	SIZE ($\times 10^{-9}$)	7.86	36.68	0.19	0.43	1.25	4.12	14.25
Book-to-market ratio	BTM	0.55	0.47	0.13	0.25	0.44	0.72	1.04
Momentum	MOM	19.02	66.62	−33.79	−12.32	9.75	34.83	70.80
Market loading	BETA	1.08	0.95	0.11	0.55	1.00	1.52	2.15
Illiquidity	ILLIQ	0.10	1.90	0.00	0.00	0.00	0.01	0.06
Maximum return (%)	RMAX	3.27	2.22	1.32	1.82	2.68	4.04	5.92

Notes: This table presents the descriptive statistics for the relevant cross-sectional variables of interest. It includes the following statistical measures for each variable: mean, standard deviation, and percentiles at the 10th, 25th, 50th (median), 75th, and 90th levels. The sample ranges from January 1994 to December 2022 and comprises all common stocks (identified as shrcd = 10, 11) traded on the NYSE, AMEX, or NASDAQ. Stocks with month-end prices below $5 per share are excluded from the analysis. The data are collected on a monthly basis. For detailed definitions of the variables, please refer to Appendix C.

Table 2
Pooled correlations.

Panel A: Corr. with DIS		Panel B: Correlations among disagreement proxies			
			AFD	IVOL	SIR
SIR	0.12	AFD	1.00		
IVOL	0.47	IVOL	0.08	1.00	
ISKEW	0.51	SIR	0.07	0.14	1.00
AFD	0.10				
SIZE	−0.05				
BTM	0.08				
MOM	−0.05				
BETA	0.12				
ILLIQ	0.04				
RMAX	0.46				

Notes: Panel A presents pooled correlations between our disagreement measure (DIS) and stock characteristics. Panel B presents pooled correlations among major disagreement proxies, AFD, IVOL, and SIR. The sample ranges from January 1994 to December 2022 and comprises all common stocks (identified as shrcd = 10, 11) traded on the NYSE, AMEX, or NASDAQ. For detailed definitions of the variables, please refer to Table 1 and Appendix C.

Table 3
Portfolio sorts on disagreement, DIS.

DIS	Raw return	Excess return	CAPM-α	FF3-α	FFC-α
Low	0.97	0.79	0.28	0.26	0.19
2	0.97	0.79	0.19	0.16	0.15
3	0.93	0.75	0.06	0.05	0.05
4	0.84	0.66	−0.10	−0.11	−0.07
5	0.79	0.61	−0.20	−0.20	−0.16
6	0.93	0.75	−0.11	−0.13	−0.02
7	0.66	0.48	−0.45	−0.46	−0.31
8	0.65	0.47	−0.55	−0.55	−0.36
9	0.30	0.12	−1.03	−0.99	−0.78
High	−0.04	−0.22	−1.49	−1.43	−1.23
Low–High	1.01	1.01	1.78	1.69	1.42
t-statistic	(2.09)	(2.09)	(4.95)	(5.53)	(4.41)
p-value	0.038	0.038	<0.001	<0.001	<0.001

Notes: This table reports the mean values of value-weighted monthly returns at month $t+1$, based on univariate portfolios sorted according to disagreement (DIS) at date t, excluding firm-months with negative ISKEW to match the theoretical prediction of Proposition 2 (which applies to positively skewed stocks). Table I.2 in the Internet Appendix shows that results are robust to including all stocks. The sample ranges from January 1994 to December 2022. The row labeled "Low–High" presents the differences in monthly returns between the 1st decile (representing low disagreement) and the 10th decile (representing high disagreement). "Raw returns" refer to one-month-ahead monthly returns, whereas "Excess returns" are defined as one-month-ahead monthly returns minus the risk-free rate. The alphas (CAPM-α, FF3-α, FFC-α) represent intercepts from a time-series regression of monthly excess returns at month $t+1$ against the market return (CAPM), further adjusted for size and value factors (FF3), and additionally a momentum factor (FFC) at month $t+1$. t-statistics, adjusted according to Newey and West (1987), are presented in parentheses along with their corresponding p-values.

5. Empirical results

In this section, we develop eight sets of analyses that are not independent findings but interlocking tests of a single theoretical framework. The logical progression is deliberate: Sections 5.1–5.2 show that DIS significantly predicts returns; Section 5.3 shows that DIS largely subsumes the predictability of IVOL, the individual proxy most closely linked to latent disagreement in the model; Section 5.4 tests whether DIS's signal is genuinely idiosyncratic, as the multi-asset separation of Appendix B implies; Section 5.5 provides support for the economic mechanism through which DIS operates, distinguishing it from the friction-based channel; Section 5.6 presents results consistent with the implication that the model's nonlinear structure is essential to these results; Sections 5.7 and 5.8 then characterize the conditions when DIS detects disagreement that is undetected by extant disagreement measures and the economic content of what DIS captures.

5.1. Portfolio sorts: Disagreement and return predictability

To examine the link between disagreement and future returns, we perform portfolio sorts, reported in Table 3. Stocks are categorized monthly into deciles based on their level of disagreement, DIS, excluding firm-months with negative ISKEW to be consistent with Proposition 2, which predicts a negative disagreement–expected return relationship for positively skewed stocks. Because the vast majority of stocks are positively skewed, as reported in Table 1, our inferences are robust to the inclusion of negative ISKEW stocks and we include all stocks in subsequent analyses unless stated otherwise. See Table I.2 in the Internet Appendix for the version of Table 3 that includes all stocks. For each decile, value-weighted one-month-ahead returns are computed via five measures: raw returns, excess returns (stock i return minus risk-free rate), CAPM-α (market factor), FF3-α (adding size/value factors), and FFC-α (adding momentum per Carhart, 1997). Value-weighted decile portfolio returns per metric are calculated, with significance reported. Returns and alphas generally decrease across rising DIS deciles, though not monotonically, with the decline steepening in the upper deciles. All L–H long-short portfolios show positive, significant annualized alphas at the 1% level or better: 21.3% CAPM-α (t = 4.95), 20.3% FF3-α (t = 5.53), and 17.0% FFC-α (t = 4.41). We note that p-values throughout are unadjusted for multiple tests and should be interpreted with appropriate caution, though the empirical work is based on implications from a theoretical model, which should be treated differently than an unstructured data mining approach (Harvey, 2017). That said, all three alpha t-statistics exceed the multiple-testing-adjusted threshold of 3.0 proposed by Harvey et al. (2016). Although it is standard empirical practice to control for factor premia, note that our theoretical framework explains why these alphas should survive standard factor risk adjustments (Appendix B).

Alphas are primarily driven by the short leg: the high-DIS decile generates −1.49% monthly CAPM alpha, while the low-DIS decile generates +0.28% (though statistically significant, t-stat. 2.88). The ratio of short-leg to long-leg contribution is approximately 5:1. This behavior is a nearly universal feature of anomaly portfolios that has been attributed to sentiment-driven overpricing (Stambaugh et al., 2012) and, relatedly, to probability-weighted demand for lottery-like payoffs (Barberis and Huang, 2008). Our model provides a complementary micro-foundation under standard expected utility preferences: for assets with positive idiosyncratic skewness (almost all stocks), disagreement generates overpricing, producing short-leg dominance through a mechanism that operates independently of both aggregate sentiment and probability weighting.

Notably, shorting is feasible and cost-effective for our sample's stocks. D'Avolio (2002) documents low shorting costs for most U.S. stocks (91% with annual fees <1%, value-weighted mean of 17 bps), with higher costs concentrated in small, illiquid names. Nagel (2005) uses institutional ownership (IO) as a short-sale constraint proxy, finding stronger return predictability in low-IO stocks; excluding the bottom 20% of IO yields weaker anomalies but easier shorting. Engelberg et al. (2018) note elevated costs and risks of maintaining short positions in the smallest size quintile. Our sample's restrictions (2+ analysts, price > $5) already omit the smallest, most illiquid, retail-heavy stocks, yielding relatively large stocks (average market cap 63% larger than unrestricted ones; the average size in our 10th DIS decile is larger than about 60% of unrestricted stocks). Moreover, over 99% of stocks in our sample have positive short interest, indicating that they are indeed shortable.

Further excluding the union of the 20% most illiquid and 20% most retail-owned stocks on breakpoints from the CRSP-COMPUSTAT universe (price > $5) removes only 9.4% of our observations (vs. 27.9% in the broader universe). DIS-decile spread alphas are robust to this exclusion: e.g., annualized CAPM-α stays at 21.5% (t-stat. 4.95), average ILLIQ in the 10th DIS decile drops from 0.46 (median 0.02) to <0.07 (median < 0.01), and average IO rises from 0.55 to 0.65. See Table I.3 in the Internet Appendix for other alphas under this exclusion, comparable to Table 3, and Tables I.9 and I.10 for average DIS-decile characteristics, with and without the exclusion. Because the spread portfolio is value-weighted, but the reported ILLIQ is an equal-weighted average that skews higher, effective illiquidity is lower than these already low reported averages (median ILLIQ 0.02 in the 10th DIS decile). DIS portfolios also exhibit relatively low turnover due to their value-weighted construction and the persistence of DIS decile assignments, especially in the extreme deciles (Table I.11 in the Internet Appendix).

Given our sample's composition, the alphas are economically large in magnitude, among the largest documented for disagreement proxies. For instance, Bali et al. (2026) employ a machine-learning-based disagreement measure on a broader sample that includes smaller and more illiquid stocks. They report the highest annualized CAPM-α among extant disagreement measures: about 17%. Direct comparison requires caution given sample differences. For example, their sample is broader and includes substantially less liquid stocks. Further, our disagreement measure is also much more persistent, implying lower turnover. That DIS generates alphas of comparable or greater magnitude on a sample that excludes the stocks where anomaly profits are typically most concentrated is consistent with the model isolating disagreement-related return variation rather than liquidity-related return patterns. Compared to other proxies evaluated on our sample, our CAPM-α exceeds that of an equivalent analyst forecast dispersion-sorted portfolio by 8.9%, an IVOL decile spread by 6.1%, and a short-interest decile spread by 13.1% (Table I.4 in the Internet Appendix).

The magnitude and persistence of these alphas raise a natural question: why does arbitrage not eliminate the overpricing that our model predicts? Our framework provides an answer that does not appeal to exogenous frictions (formally tested in Section 5.5). In the standard limits-of-arbitrage framework (Shleifer and Vishny, 1997; Pontiff, 2006), mispricing persists because arbitrageurs face costs or risks that prevent them from fully correcting it. Our model identifies a distinct mechanism. Even unconstrained arbitrageurs who short overpriced, high-disagreement stocks effectively hold a position in a negatively skewed asset, because the short side of a positively skewed stock is itself negatively skewed. The demand convexity documented in Section 2.4.1 implies that such positions are unattractive relative to their expected payoff because the same prudence motive that generates overpricing in equilibrium also discourages the arbitrage trades that would correct it. This is not a friction-based limit but a preference-based one. Rational, unconstrained agents with standard preferences optimally choose smaller short positions than the magnitude of mispricing would warrant under symmetric payoffs. The result is that disagreement-driven overpricing in positively skewed stocks is partially self-insulating, and the alphas documented above are a prediction of the model.

Table 4
Fama and MacBeth (1973) cross-sectional regressions.

	(1) $EXRET_{t+1}$	(2) $EXRET_{t+1}$	(3) $EXRET_{t+1}$	(4) $EXRET_{t+1}$
DIS	−1.55***	−0.97***	−1.36***	−0.96***
	(−3.98)	(−3.23)	(−4.59)	(−3.27)
SIR		−3.75***		−3.45**
		(−2.64)		(−2.46)
IVOL		−0.04		−0.05
		(−0.56)		(−0.94)
ISKEW		−0.38**		−0.23
		(−2.16)		(−1.39)
AFD		−0.16**		−0.14**
		(−2.22)		(−2.16)
LnSIZE			−0.04	−0.09**
			(−1.00)	(−2.08)
BTM			0.09	0.06
			(0.58)	(0.45)
MOM			0.34	0.30
			(1.62)	(1.41)
BETA			−0.07	−0.03
			(−0.55)	(−0.26)
ILLIQ			−0.47	−0.02
			(−1.33)	(−0.07)
Constant	0.82***	1.29***	0.98**	1.71***
	(2.94)	(4.91)	(2.07)	(3.26)
Observations	719,940	719,940	701,159	701,159

Notes: This table displays the average coefficients and t-statistics (in parentheses) from a monthly Fama and MacBeth (1973) regression analysis conducted on excess returns, adhering to the methodologies in Fama and French (1993) and Carhart (1997). These regressions are performed against a lagged measure of disagreement and various lagged stock and firm characteristics. The sample ranges from January 1994 to December 2022. The coefficients on DIS, AFD, and MOM are multiplied by 100 for readability. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Appendix C.

5.2. *Fama and MacBeth (1973) cross-sectional regressions*

Next, we perform cross-sectional predictive regressions of excess returns (EXRET) at month $t+1$ in the spirit of Fama and MacBeth (1973). Table 4 reports regression coefficients with Newey and West (1987) adjusted t-stats in parentheses. Column (1) reveals a negative DIS impact on future EXRET (coeff. −1.55, t-stat. −3.98). Because DIS coefficients are reported multiplied by 100, the predicted monthly excess return spread equals the DIS spread times the reported coefficient divided by 100. For instance, from Table 10, the DIS spread between high and low deciles (66.67 = 67.67−1.00) multiplied by the DIS coefficient (−1.55/100) yields a predicted monthly return spread of approximately 1.03% (or 12.4% annualized), aligning with the decile-portfolio excess return spread in Table 3.

Column (2) shows that this negative relation holds after controlling for DIS's major components, with DIS still predicting lower returns with the coefficient somewhat reduced but still over three standard errors from zero, despite relatively high correlations with IVOL and ISKEW. DIS absorbs most of IVOL's negative variation, while ISKEW, AFD, and SIR remain negative predictors. We explore the relationship between DIS and IVOL in greater detail in Section 5.3.

Columns (3) and (4) add established return predictors (ILLIQ, LnSIZE, BTM, MOM, BETA) and then combine these with DIS's components; DIS's predictive power persists across both specifications with t-statistics exceeding three in magnitude. The results imply DIS's marginal contribution to a low-minus-high decile spread is 7.7% annualized, net of other cross-sectional determinants including extant disagreement proxies.

5.3. DIS and the idiosyncratic volatility puzzle

Given the evidence that DIS predicts returns in the cross section (Sections 5.1–5.2), we now examine its relationship to the individual proxy most closely linked to latent disagreement in our model, idiosyncratic volatility. Ang et al. (2006) document a strong negative relation between IVOL and subsequent stock returns, which has proven difficult to rationalize. A substantial literature has sought explanations for the IVOL effect; Hou and Loh (2016) offer a comprehensive review, concluding that no approach fully resolves the puzzle. In our full sample, which comprises larger and more liquid stocks than those typically studied in the IVOL literature, we find a significant (at the 5% level) negative bivariate relation between IVOL and subsequent returns. As noted above, when DIS is included in the regression, it absorbs much of the variation.

To further investigate DIS's absorption of IVOL predictability, we examine smaller stocks in our sample using monthly SIZE breakpoints from the broader CRSP/COMPUSTAT universe (price > $5). Lower breakpoints yield more restrictive subsamples: e.g., below the midpoint corresponds to our sample's smallest 30.3%, and below the 25th percentile to the smallest 7.9%. Table 5 reports regressions for DIS and IVOL, individually and jointly, across four subsamples: full, smallest half, third, and quarter. As size decreases, IVOL's individual predictive strength grows (t-stat. from −2.14 full-sample to −4.70 in the smallest quarter), consistent with prior work. DIS shows a similar pattern. Yet, even as IVOL's effects strengthen in smaller stocks, DIS reliably absorbs its effects in that IVOL is insignificant in all size samples when DIS is included in the regression.

In Section 5.4, we further discuss how DIS reliably absorbs IVOL's predictability across different macro regimes. In Section 5.5, we document that IVOL's return predictability is also friction-dependent, whereas DIS's predictability appears to be robust to frictions, identifying another source of contaminating variation in IVOL that DIS's equilibrium structure appears to filter out. Tests in Section 5.7 show how DIS captures additional disagreement effects beyond what is detected by IVOL.

These results suggest a disagreement-based explanation for IVOL's return predictability. DIS's nonlinear equilibrium structure appears to be able to extract the disagreement-related component of IVOL's signal, while filtering the regime-dependent and friction-dependent components that contaminate IVOL as a standalone measure (Sections 5.4–5.5). Our theoretical model, underpinning DIS, links IVOL closely to disagreement, incorporating it prominently in our composite proxy. Together, these findings provide a micro-founded view of IVOL's standalone effects arising in large part as an investor disagreement correlate. Relatedly, Bali et al. (2026) report that their machine-learning-based disagreement measure is also most highly correlated with IVOL among reported correlations, consistent with our model's prediction that any well-specified measure of asset-specific disagreement should load heavily on idiosyncratic volatility.

Table 5
Fama and MacBeth (1973) cross-sectional regressions with DIS and IVOL for subsamples based on CRSP/COMPUSTAT market size thresholds.

Sample	DIS		IVOL		N	N%
Full sample	−1.55***	(−3.98)			719,940	100.0%
			−0.16**	(−2.14)		
	−1.26***	(−3.64)	−0.08	(−1.00)		
Smallest half	−1.51***	(−4.65)			218,352	30.3%
			−0.16**	(−2.45)		
	−1.56***	(−4.35)	0.01	(0.18)		
Smallest third	−1.46***	(−5.03)			101,765	14.1%
			−0.23***	(−3.62)		
	−1.18***	(−2.96)	−0.06	(−0.72)		
Smallest quarter	−1.74***	(−5.58)			56,912	7.9%
			−0.32***	(−4.70)		
	−1.47***	(−3.01)	−0.08	(−0.85)		

Notes: This table displays the average coefficients and t-statistics (in parentheses) from a monthly Fama and MacBeth (1973) regression analysis conducted on excess returns, adhering to the methodologies in Fama and French (1993) and Carhart (1997). These regressions are performed against a lagged composite disagreement DIS, lagged IVOL, and a constant (not reported) for various subsamples based on monthly SIZE breakpoints from common stocks in CRSP/COMPUSTAT having price above $5. N denotes the number of observations and N% indicates the observations count relative to our original sample that includes all stocks with 2+ analysts for which DIS is measurable. The sample ranges from January 1994 to December 2022. The coefficients on DIS are multiplied by 100 for readability. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Appendix C.

5.4. Investor disagreement and macroeconomic regimes

The absorption result of Section 5.3 raises a natural question: is DIS's predictive power genuinely idiosyncratic, or does it partly reflect aggregate variation that may contaminate the information in both DIS and IVOL? A key question for interpreting disagreement measures is whether they capture genuinely idiosyncratic, firm-level variation or instead reflect aggregate phenomena such as macroeconomic uncertainty or market-wide volatility, in which case their return predictability could partly reflect compensation for common risk rather than the firm-specific overpricing mechanism our model describes. Indeed, Bigio et al. (2026) show that heterogeneous beliefs can themselves directly drive aggregate fluctuations in asset prices and real activity, underscoring the need to separate aggregate belief dynamics from the firm-level channel. The multi-asset separation result of Appendix B sharpens this into a testable restriction: a well-specified idiosyncratic disagreement measure should predict returns independently of the macroeconomic regime, since aggregate variation is absorbed into factor premia and does not enter the idiosyncratic pricing channel. We show that DIS is consistent with this restriction, but the individual proxies are not.

We split the sample into high and low states based on the median of eight aggregate indicators: the CBOE VIX Index; Volatility-of-Volatility, the realized volatility of daily VIX changes over the prior month (Bekaert and Hoerova, 2014); the CATFIN systemic risk measure of Allen et al. (2012); the Financial, Real, and Macroeconomic Uncertainty indices of Jurado et al. (2015); the partial-least-squares Sentiment Index of Kelly and Pruitt (2013); and Aggregate IVOL, the value-weighted average of stock-level idiosyncratic volatilities. Within each regime, we estimate Fama and MacBeth (1973) regressions of one-month-ahead excess returns on each disagreement proxy with controls (BETA, LnSIZE, BTM, MOM). Table 6 reports the coefficient and t-statistic for each proxy (DIS, IVOL, AFD, SIR).

DIS is a significant negative predictor at the 5% level or better in both high and low states across all eight indicators (16 of 16 total states), with coefficients broadly stable across regimes. The individual proxies are less consistent. IVOL is significant at the 5% level or better in low states for 7/8 indicators, but only 1/8 high-state subsamples (8 of 16 total states). SIR is significant at the 5% level or better in 6/8 low-state subsamples, but only 1/8 high-state subsamples (7 of 16 total

Table 6
Return predictability in High or Low macroeconomic regimes.

State	DIS		IVOL		AFD		SIR	
Macro indicator	Coeff.	*t*-stat.	Coeff.	*t*-stat.	Coeff.	*t*-stat.	Coeff.	*t*-stat.
High								
VIX Index	−1.16**	(−2.54)	−0.07	(−0.80)	−0.26*	(−1.90)	−1.33	(−0.46)
Volatility-of-Volatility	−0.67**	(−2.38)	−0.10	(−1.33)	−0.27**	(−2.27)	−3.37*	(−1.80)
CATFIN	−1.40***	(−3.66)	−0.19**	(−2.31)	−0.39***	(−3.04)	−5.34***	(−2.80)
Financial Uncertainty	−0.90**	(−2.37)	−0.13	(−1.35)	−0.28**	(−2.22)	−2.60	(−0.86)
Real Uncertainty	−0.99***	(−3.30)	−0.09	(−1.13)	−0.23**	(−2.48)	−0.78	(−0.27)
Macroeconomic Uncertainty	−1.34***	(−3.54)	−0.15	(−1.65)	−0.28**	(−2.45)	−2.01	(−0.70)
PLS Sentiment Index	−1.35***	(−2.95)	−0.12	(−1.26)	−0.28*	(−1.92)	−2.66	(−0.92)
Aggregate IVOL	−1.62***	(−3.37)	−0.15	(−1.54)	−0.32**	(−2.15)	−3.82	(−1.25)
Low								
VIX Index	−1.60***	(−4.29)	−0.22***	(−3.99)	−0.22**	(−2.49)	−6.45***	(−5.42)
Volatility-of-Volatility	−2.09***	(−4.70)	−0.20***	(−2.62)	−0.21*	(−1.95)	−4.41*	(−1.88)
CATFIN	−1.36***	(−3.25)	−0.10	(−1.39)	−0.09	(−0.96)	−2.43	(−0.96)
Financial Uncertainty	−1.86***	(−4.49)	−0.16***	(−3.27)	−0.20**	(−2.03)	−5.18***	(−4.60)
Real Uncertainty	−1.77***	(−3.48)	−0.20**	(−2.47)	−0.25*	(−1.79)	−6.99***	(−5.00)
Macroeconomic Uncertainty	−1.42***	(−3.10)	−0.15**	(−2.30)	−0.20*	(−1.69)	−5.77***	(−4.17)
PLS Sentiment Index	−1.54***	(−3.60)	−0.15**	(−2.10)	−0.20**	(−2.05)	−3.81**	(−2.34)
Aggregate IVOL	−1.14***	(−3.67)	−0.14***	(−3.10)	−0.16**	(−2.46)	−3.95***	(−4.08)

Notes: This table displays the average coefficient and *t*-statistic (in parentheses) of each disagreement proxy (DIS, IVOL, AFD, or SIR; in columns) from a monthly Fama and MacBeth (1973) regression analysis performed on excess returns against the disagreement proxy, standard cross-sectional controls (BETA, LnSIZE, BTM, and MOM), and a constant (not reported). Regressions are performed for various subsamples based on median breakpoints of eight macro indicators: VIX; Volatility-of-Volatility following Bekaert and Hoerova (2014); CATFIN, the systemic risk measure of Allen et al. (2012); the Financial Uncertainty, Real Uncertainty, and Macroeconomic Uncertainty indices of Jurado et al. (2015); the PLS Sentiment Index of Kelly and Pruitt (2013); and Aggregate IVOL. For each indicator, high (low) states are defined as months in which the indicator is above (below) its full-sample median. The coefficients on DIS and AFD are multiplied by 100 for readability. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Appendix C.

states). AFD is more consistent, with significance at the 5% level or better across 6/8 high states and 4/8 low states (10 of 16 total states), though at uniformly less significance than DIS.

In further tests, regressing excess returns jointly on DIS, IVOL, and controls within each macro regime reveals that DIS absorbs or weakens IVOL's predictability in all high and low regimes except the VIX low state (see Table I.7 in Internet Appendix I), extending the absorption result of Section 5.3 to the time-series dimension, but through two distinct mechanisms. In high-uncertainty states, IVOL is already non-predictive on its own, so its insignificance in the joint regression reflects a lack of independent content. In low-uncertainty states, IVOL carries a meaningful standalone signal that is nonetheless largely subsumed by DIS when the two enter jointly. This decomposition refines the potential micro-founded resolution of the IVOL puzzle offered in Section 5.3: IVOL's predictability is the regime-dependent portion of a broader disagreement effect, present when aggregate noise is low enough for IVOL to proxy reasonably for latent disagreement, and absent when aggregate volatility contamination severs that link.

The pattern follows from DIS's multiplicative form. In periods of aggregate stress, idiosyncratic volatility and short interest rise broadly across the cross section, compressing the informative variation among stocks and making it difficult to distinguish genuinely high-disagreement firms from those whose values are elevated for macro-related reasons. The cross-sectional predictive slope degrades because the *ranking* becomes contaminated. DIS's structure provides insulation by requiring corroboration across components: a stock's position in the cross-sectional ranking is driven by the joint configuration of IVOL, SIR, and AFD rather than by any single component. Because analyst forecast dispersion is driven primarily by firm-specific information uncertainty and is less sensitive to market-wide volatility, this co-occurrence requirement preserves the cross-sectional variation that distinguishes genuinely high-disagreement firms even when aggregate conditions inflate the individual components broadly.

These results can help explain the conflicting empirical findings across samples using different proxies. Each proxy captures a partial, macro-uncertainty-contaminated slice of latent disagreement whose predictive content varies with the macro environment of the sample period. A researcher using IVOL in a low-volatility period will reach systematically different conclusions than one using SIR in a high-uncertainty period, not because the underlying phenomenon differs, but because the proxies are contaminated in ways that degrade their cross-sectional predictive content, whereas DIS's multi-component structure appears to mitigate this degradation.

Our use of regime-conditioning is the contrapositive logic of prior work such as that of Stambaugh et al. (2012), who test whether a proxy's predictive power varies with regimes in a theoretically predicted direction. We instead assess whether it does not vary. This diagnostic principle may prove useful in other settings where researchers must distinguish idiosyncratic from aggregate signals using imperfect proxies. An empirical proxy that purports to capture a firm-specific, idiosyncratic phenomenon should, under the logic of factor-structure separation, exhibit return predictability that is consistent across aggregate macroeconomic states.

5.5. The role of short-selling frictions

Sections 5.3 and 5.4 have provided evidence suggesting that DIS largely subsumes IVOL's predictability and behaves consistently with a measure of idiosyncratic disagreement. We now ask through what economic mechanism DIS's predictive power operates, and in particular whether it reflects the frictionless skewness channel identified by our model or the friction-based channel of Miller (1977). Local demand convexity can arise from two sources: positive skewness, as in our model (Section 2.4.1), or short-sale frictions. For example, a prohibition on short sales would produce a kink in demand at the investor's belief, generating local convexity even without skewness. Since Miller (1977), the dominant explanation for disagreement-driven overpricing has relied on the latter source: short-sale constraints prevent pessimists from fully expressing their views, biasing prices toward optimists' valuations. Hong and Sraer (2016) show that overpricing can arise even without binding constraints if bearish investors simply hold zero rather than negative positions, though the mechanism still relies on the absence of short positions. Our model operates via a distinct channel in which the combination of heterogeneous beliefs and positive payoff skewness is sufficient to generate overpricing even when investors do short. They simply short less aggressively than optimists buy, because the short side

Table 7
Marginal effect of measure D as institutional ownership approaches 100%: Fama and MacBeth (1973) cross-sectional interaction regressions.

	D = DIS		D = IVOL		D = AFD		D = SIR	
β_D	−2.46***	−1.98***	−0.00	0.00	0.01	0.03	−0.53	−0.51
	(−3.12)	(−3.09)	(−0.00)	(0.02)	(0.07)	(0.24)	(−0.31)	(−0.32)
β_{1-IO}	−0.12	−0.18	0.70***	0.65***	−0.44**	−0.39**	−0.17	−0.12
	(−0.64)	(−0.98)	(2.72)	(2.68)	(−2.12)	(−2.03)	(−0.67)	(−0.52)
$\beta_{D\times(1-IO)}$	0.81	0.31	−0.41***	−0.40***	−0.52	−0.44	−15.84***	−13.99***
	(0.92)	(0.39)	(−4.23)	(−4.16)	(−1.61)	(−1.61)	(−4.05)	(−3.99)
Controls	No	Yes	No	Yes	No	Yes	No	Yes

Notes: This table reports average coefficients for main effects and interaction terms with t-statistics (in parentheses) from Fama and MacBeth (1973) cross-sectional interaction regressions of subsequent excess returns on disagreement measure D (among DIS, IVOL, AFD, SIR), non-institutional ownership $1-IO$, and their interaction $D\times(1-IO)$, first unconditionally (no controls) in the left-hand column for each measure and second conditionally on a vector of standard cross-sectional control variables Z in the right-hand column,

$$EXRET_{t+1} = \alpha + \beta_D \cdot D + \beta_{1-IO} \cdot (1-IO) + \beta_{D\times(1-IO)} \cdot (D\times(1-IO)) + \beta_Z' Z + \varepsilon,$$

where Z includes BETA, LnSIZE, BTM, and MOM, and β_Z is their corresponding coefficient vector; and where ε is an error term. The coefficient β_D represents the marginal effect of measure D when institutional ownership is 100% ($\beta_D = \frac{\partial EXRET_{t+1}}{\partial D}$ as $IO \nearrow 1$). The sample ranges from January 1994 to December 2022. Coefficients on DIS, AFD, and their interactions are multiplied by 100 for readability. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Appendix C.

of a positively skewed asset is itself negatively skewed. Both channels may contribute in practice. The question we address in this section is whether the frictionless channel is empirically operative, and whether the two channels can be distinguished in the data.

We follow Nagel (2005) in using institutional ownership proportion (IO) as an inverse proxy for short-selling frictions and estimate the Fama and MacBeth (1973) interaction regression

$$EXRET_{t+1} = \alpha + \beta_D \cdot D + \beta_{1-IO} \cdot (1-IO) + \beta_{D\times(1-IO)} \cdot (D\times(1-IO)) + \beta_Z' Z + \varepsilon, \tag{19}$$

for each disagreement measure D among DIS, IVOL, AFD, and SIR, where Z includes BETA, LnSIZE, BTM, and MOM. Because the data are not demeaned and the interaction term is included, β_D is correctly interpreted as the marginal effect of D as institutional ownership approaches 100% (Balli and Sørensen, 2013). Stocks with full or near-full institutional ownership are those for which locating shares to borrow is least costly and short-selling frictions are minimally binding. If the disagreement–return relationship operated primarily through constraints on short-selling, the marginal effect of DIS should be economically and statistically attenuated among these stocks.

This specification nests both channels and generates distinct empirical predictions for each. Under the friction-based channel alone, the disagreement–return relationship should intensify as frictions increase: the interaction coefficient $\beta_{D\times(1-IO)}$ should be significantly negative, and the level effect as institutional ownership approaches 100%, β_D, should be zero, since the mechanism has no force when constraints are absent. Under the frictionless skewness channel, the relationship should be present even approaching full institutional ownership: β_D should be significantly negative, and the interaction should be zero or at least not necessary to generate the effect. The two channels are not mutually exclusive; both could appear simultaneously in the same measure. The empirical question is which pattern each measure exhibits.

Table 7 reports the results for all four measures. DIS is the only measure with a statistically significant marginal effect as institutional ownership approaches 100%: −2.46 (t = −3.12) unconditionally and −1.98 (t = −3.09) with standard cross-sectional controls, both significant at the 1% level. These magnitudes are comparable to the unconditional Fama–MacBeth results of Table 4, suggesting that the predictive content of DIS is not attenuated as short-selling frictions diminish. The interaction of DIS with $(1-IO)$ is statistically indistinguishable from zero in both specifications. This pattern is consistent with the frictionless skewness channel, for which the disagreement–return relationship operates through demand asymmetry that does not depend on constrained short-selling.

The individual proxies exhibit a different pattern. Neither IVOL, AFD, nor SIR generates a significant marginal effect as institutional ownership approaches 100%. The level coefficients are economically negligible and statistically insignificant across all specifications. In contrast, IVOL and SIR exhibit highly significant interactions with $(1-IO)$. These patterns are consistent with the friction-based channel. The return predictability of these proxies is concentrated in stocks where short-selling constraints are more binding and vanishes where frictions are absent. AFD exhibits a marginally insignificant interaction, consistent with its distinct role in our model as an amplifier of existing disagreement, a public information effect, rather than a direct measure of the trading-based mechanism through which frictions operate.

Two channels of disagreement pricing. Taken together, the results in Table 7 provide evidence that disagreement generates overpricing through at least two distinct channels that coexist in the data and are empirically separable. These contrasting patterns emerge from a common specification applied to the same sample, providing identification that would not be available from studying any single measure in isolation. This identification is as clean as the institutional-ownership proxy permits: IO approaching 100% does not literally eliminate all shorting frictions, but represents the limiting case of the standard proxy in this literature (Nagel, 2005), and the contrasting patterns across measures within this common framework provide the empirical separation. The two channels are complementary rather than competing. The results suggest that IVOL and SIR each capture a partial slice of latent disagreement that is entangled with the friction mechanism, while DIS, whose nonlinear multi-component structure filters these observables, exhibits the predictive behavior the model associates with the frictionless disagreement channel. DIS generating alphas exceeding those of any individual proxy on the same sample (Section 5.1) is consistent with this interpretation as it captures a channel that is operative broadly, rather than one that is active only conditionally.

The cross-sectional evidence of this section complements the macro-regime evidence of Section 5.4 in that periods of elevated aggregate uncertainty are also periods when shorting costs tend to rise (cf. Engelberg et al., 2018), yet DIS predicts returns consistently across these states while the three standalone proxies do not. That DIS is supported by both tests, friction independence and regime stability, while IVOL and SIR are not, provides dual identification support that DIS captures a fundamentally different channel from its individual components. The union of both dimensions points to the skewness-based demand asymmetry identified by the model as the operative mechanism. This refines the limits-of-arbitrage interpretation of disagreement anomalies. The binding constraint is not that arbitrageurs cannot short overpriced stocks, but that shorting a positively skewed stock exposes the arbitrageur to negatively skewed payoffs, reducing optimal position sizes below what would be needed to eliminate the mispricing. Frictions, where present, provide an additional source of overpricing, and the

individual proxies' sensitivity to IO variation confirms that this additional source is real. In sum, our analysis provides new evidence that the frictionless channel operates alongside the friction-based channel.

Joint implications for the IVOL puzzle. Sections 5.3 through 5.5 jointly characterize the relationship between IVOL and latent disagreement. Our model links IVOL closely to the latent variable, and DIS largely absorbs IVOL's return predictability even in the smaller-stock subsamples where that predictability is strongest (Section 5.3). Yet IVOL's standalone predictability is both regime-dependent, present in low-uncertainty states but absent in high-uncertainty states where aggregate volatility contaminates its signal (Section 5.4), and friction-dependent, concentrated where short-selling constraints bind (Section 5.5). DIS exhibits neither dependence. The complete picture is that IVOL inherits its negative relation with future returns from its proximity to latent disagreement, but that predictability is unreliable because IVOL captures one margin of a nonlinear multi-component latent variable while its predictive content remains exposed to aggregate and friction-related variation that DIS's equilibrium structure appears to filter out. We note that these results constitute a theory-grounded account rather than a formal causal decomposition; however, the conjunction of absorption, regime diagnosis, and friction diagnosis, each predicted by the model's characterization of IVOL's relationship to latent disagreement, provides evidence that is cumulative in a way not available to prior proposed explanations. In particular, Stambaugh et al. (2015) attribute the IVOL puzzle to arbitrage asymmetry in that overpricing is harder to correct than underpricing because shorting is costlier than buying, so high-IVOL stocks are where mispricing persists. Our account is complementary but distinct because the asymmetry in our model arises from preferences rather than costs, and the regime and friction diagnoses identify aggregate and friction-related variation in IVOL that an arbitrage-asymmetry framework, which relies on the same shorting frictions, does not address.

Table 8
Fama and MacBeth (1973) cross-sectional regressions by DIS and linear composite disagreement alternatives.

	(1) $EXRET_{t+1}$	(2) $EXRET_{t+1}$	(3) $EXRET_{t+1}$	(4) $EXRET_{t+1}$	(5) $EXRET_{t+1}$
DIS	−1.55***	−1.20***	−1.54***	−1.01***	−1.70***
	(−3.98)	(−4.31)	(−4.59)	(−3.68)	(−4.83)
DEW		−0.08			
		(−1.39)			
DEW+			−0.04		
			(−0.79)		
DPCA				−0.20*	
				(−1.80)	
DPCA+					−0.02
					(−0.30)
Constant	0.82***	1.25***	1.05***	0.74***	0.80***
	(2.94)	(4.55)	(3.64)	(2.59)	(2.84)
Observations	719,940	719,940	717,344	719,940	717,344

Notes: This table displays the average coefficients and t-statistics (in parentheses) from a monthly Fama and MacBeth (1973) regression analysis conducted on excess returns, adhering to the methodologies in Fama and French (1993) and Carhart (1997). These regressions are performed against a lagged measure of disagreement and lagged alternative linear disagreement measures. The sample ranges from January 1994 to December 2022. DEW is a linear disagreement score constructed as follows: in each month, assign each stock to a decile based on each of AFD, IVOL, and SIR, and then take a simple average of these decile ranks. DEW+ is constructed similarly, except deciles are formed on AFD, IVOL, and ASIR, as well as four disagreement variables BIDASK, SUV, DTO, and DBREADTH. DPCA is the first component of a principal component analysis based on AFD, IVOL, and SIR. DPCA+ is the first component of a principal component analysis based on AFD, IVOL, ASIR, BIDASK, SUV, DTO, and DBREADTH. The specifications including DEW+ and DPCA+ have fewer observations because the additional variables (BIDASK, SUV, DTO, and DBREADTH) have more limited data availability than the three core proxies. The coefficients on DIS are multiplied by 100 for readability. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Appendix C.

5.6. Role of nonlinearity

The preceding sections have provided evidence that DIS's predictive power is both regime-stable and friction-independent, properties not shared by individual proxies and, as we now show, not absorbed by linear combinations of the same inputs. The composite disagreement measure DIS is a nonlinear function of its constituent proxies, with the functional form guided by the model's equilibrium conditions rather than selected from a flexible family of candidate specifications. As Eq. (15) makes explicit, the components interact multiplicatively. Idiosyncratic skewness scales the impact of idiosyncratic volatility on the equilibrium price bias, while analyst forecast dispersion amplifies the scale of existing heterogeneity rather than contributing additively. DIS therefore embeds a co-occurrence requirement: a stock registers high disagreement only when multiple inputs are simultaneously elevated in proportions governed by their model-determined interactions.

To test whether this nonlinear structure matters empirically, we construct two families of linear alternatives. The first are equally weighted composites: DEW, the average of monthly decile ranks of AFD, IVOL, and SIR; and DEW+, which extends this to seven proxies by using abnormal short interest (ASIR) in place of SIR, and by additionally incorporating the high-low bid–ask spread (BIDASK), changes in mutual fund ownership (DBREADTH), market-adjusted turnover changes (DTO), and standardized unexplained volume (SUV). The second are principal-component-based measures: DPCA, the first principal component of AFD, IVOL, and SIR (capturing 40% of joint variation); and DPCA+, its seven-proxy analog (capturing 26%). Together these span the range from the most atheoretic equal-weighting scheme to the variance-maximizing principal component approach, and the expanded versions address the concern that the result is sensitive to the choice of three input proxies.

Table 8 reports Fama and MacBeth (1973) regressions of excess returns on DIS alongside each linear alternative in turn. In every specification, DIS remains a highly significant negative predictor, with coefficients ranging from −1.01 to −1.70 and t-statistics between −3.68 and −4.83. By contrast, none of the linear alternatives achieves conventional significance: DEW and DEW+ are insignificant at any standard level, and DPCA reaches only marginal significance ($t = -1.80$) while DIS alongside it remains strongly significant ($t = -3.68$). The broader seven-proxy versions perform similarly to their three-proxy counterparts. The return predictability embedded in DIS is not recoverable by these linear combinations of the same underlying information.

The model provides an explanation of this result. A linear composite weights its inputs additively, assigning high disagreement scores whenever any subset of proxies is elevated, including when IVOL and SIR are broadly inflated by aggregate stress without corresponding elevation in AFD. DIS's multiplicative structure requires corroboration across components, filtering the aggregate-driven signals that degrade linear alternatives in high-uncertainty states. The regime-stability of DIS documented in Section 5.4 and its dominance over linear alternatives here are two manifestations of the same feature of (15). The nonlinear structure is essential to DIS's performance, not incidental to it.

5.7. Detecting disagreement undetected by extant measures

In this section, we assess the degree to which and the nature of how DIS detects disagreement not detected by extant disagreement measures (IVOL, AFD, and SIR), i.e., when the extant measure approaches zero. For instance, what disagreement-associated characteristics might DIS be picking up that IVOL misses?

In Table 9, we report the estimated marginal effect β_{DIS} of composite disagreement DIS on characteristic Y when extant disagreement measure D approaches zero detection of disagreement using the interaction regression

$$Y = \alpha + \beta_{\text{DIS}} \cdot \text{DIS} + \beta_D \cdot D + \beta_{\text{DIS}\times D} \cdot (\text{DIS} \times D) + \beta_Z' Z + \varepsilon, \tag{20}$$

Table 9
Marginal effect of DIS as extant disagreement measure D approaches zero.

Y	D = IVOL		D = AFD		D = SIR	
	DIS coeff.	t-stat.	DIS coeff.	t-stat.	DIS coeff.	t-stat.
EXRET_{t+1}	-2.384^{***}	(−3.46)	-1.531^{***}	(−4.26)	-1.074^{***}	(−3.00)
IVOL			0.044^{***}	(10.94)	0.042^{***}	(12.10)
AFD	0.784^{***}	(10.22)			0.330^{***}	(10.14)
SIR	−0.008	(−0.88)	0.017^{***}	(5.53)		
ISKEW	0.027^{***}	(13.75)	0.010^{***}	(19.01)	0.009^{***}	(19.54)
LnSIZE	-0.079^{***}	(−9.31)	-0.041^{***}	(−9.57)	-0.044^{***}	(−11.16)
BTM	0.111^{**}	(2.44)	-0.049^{*}	(−1.90)	-0.106^{***}	(−4.89)
MOM	-0.799^{***}	(−5.41)	-0.141^{**}	(−2.10)	-0.162^{***}	(−2.95)
BETA	0.014^{***}	(9.53)	0.007^{***}	(8.73)	0.006^{***}	(8.22)
ILLIQ	0.017^{***}	(4.10)	0.010^{***}	(3.28)	0.013^{***}	(3.89)
RMAX	0.019^{***}	(8.73)	0.052^{***}	(10.52)	0.050^{***}	(11.64)

Notes: This table reports the estimated marginal effect β_{DIS} of composite disagreement DIS on characteristic Y when extant disagreement measure D approaches zero ($\beta_{\text{DIS}} = \frac{\partial Y}{\partial \text{DIS}}$ as $D \searrow 0$) and its Newey and West (1987) adjusted t-statistic from the Fama and MacBeth (1973) cross-sectional interaction regression of characteristic Y against DIS, an extant disagreement measure D, and their interaction (DIS × D), as well as a vector of standard cross-sectional control variables Z,

$$Y = \alpha + \beta_{\text{DIS}} \cdot \text{DIS} + \beta_D \cdot D + \beta_{\text{DIS}\times D} \cdot (\text{DIS} \times D) + \beta_Z' Z + \varepsilon,$$

which are reported (by column) for each extant disagreement measure D among IVOL, AFD, and SIR; and (by row) for each characteristic Y among EXRET_{t+1}, IVOL, AFD, SIR, ISKEW, LnSIZE, BTM, MOM, BETA, ILLIQ, and RMAX; where Z includes BETA, LnSIZE, BTM, and MOM (control not included if same variable as Y), and β_Z is their corresponding coefficient vector; and where ε is an error term. The sample ranges from January 1994 to December 2022. Coefficients are multiplied by 100 for readability when $Y = \text{EXRET}_{t+1}$, SIR, and BTM. The significance levels are denoted as ***, **, and *, indicating significance at the 1%, 5%, and 10% levels, respectively, under a single test. For detailed definitions of the variables, please refer to Table 1 and Appendix C.

for each extant disagreement measure D among IVOL, AFD, and SIR, and for each characteristic Y among EXRET_{t+1}, IVOL, AFD, SIR, ISKEW, LnSIZE, BTM, MOM, BETA, ILLIQ, and RMAX; where Z is a vector of standard cross-sectional control variables including BETA, LnSIZE, BTM, and MOM (control not included if same variable as Y).

The marginal effect of DIS on Y is $\beta_{\text{DIS}} + \beta_{\text{DIS}\times D} \cdot D$, a linear function of the extant disagreement proxy D. Because D is not de-meaned, as D approaches zero, the marginal effect is simply β_{DIS}, which is reported in Table 9 (Balli and Sørensen, 2013).

Table 9 shows that DIS is picking up disagreement associated with specific firm characteristics that extant disagreement measures fail to detect in the limit as it goes to zero. That is, in the limiting absence of disagreement detected by an extant disagreement measure (IVOL, AFD, and SIR), DIS exhibits a consistent negative relationship with EXRET_{t+1}, LnSIZE, and MOM, and a consistent positive relationship with ISKEW, BETA, ILLIQ, and RMAX.[17] These relationships suggest that DIS detects a specific dimension of return-predictive variation consistent with disagreement, which would be undetected by each of the extant measures in the limit, that is fundamentally linked to lower future returns for relatively smaller, less trending, more positively skewed, riskier, and relatively more illiquid firms (among the relatively large and liquid firms in our sample restricted to 2+ analysts).

5.8. Characteristics of disagreement DIS

Lastly, we look at characteristics associated with DIS. Table 10 reports monthly equally weighted average characteristics within DIS deciles. Consistent with our model's predictions, we observe an increase in individual components of DIS (SIR, IVOL, ISKEW, and AFD) as DIS rises. Moreover, stocks associated with higher DIS are characterized by relatively lower market capitalization (SIZE), among the larger stocks of our dataset restricted to 2+ analysts, and are consistent with potential market undervaluation (indicated by higher BTM ratios). Stocks with higher DIS also exhibit less momentum, although the relationship is statistically weaker than those of other characteristics. The relationship between DIS and future returns implies that high levels of disagreement may signal potential turning points in momentum (Goulding et al., 2023a).

Moreover, we find that DIS aligns intuitively with the expected relationship between firm opacity and investor disagreement. Stocks with higher DIS have more systematic risk (as evidenced by higher beta values), helping to explain why DIS-spread portfolios diverge from CAPM predictions and yield large alphas in upper DIS deciles. High DIS stocks also exhibit relatively lower levels of liquidity, among the liquid stocks of our dataset. Stocks with more lottery-like behavior (RMAX; Bali et al., 2011) also associate with higher DIS. Additionally, higher-DIS firms tend to exhibit increased financial constraints, younger age, intensified R&D expenditures, greater opinion divergence among mutual funds, and reduced volumes of Google searches, both overall and specifically investment-related (Table I.9 in the Internet Appendix).

6. Conclusion

We address the disagreement of disagreement problem: the near-zero correlations among major investor disagreement proxies that make it challenging to infer how disagreement impacts prices. Our resolution proceeds through a parsimonious frictionless equilibrium model that links all major proxies to a single latent disagreement variable, a nonlinear composite disagreement measure DIS guided by the model's equilibrium interrelationships, and a comprehensive empirical program that employs DIS as the vehicle through which the model's predictions are tested. The empirical evidence is cumulative and mutually reinforcing, consistent with the model's predictions about cross-sectional predictability, proxy absorption, regime invariance, friction independence, and nonlinear necessity, with each test providing identification that is distinct from the others.

Empirically, our tests are broadly consistent with the model's predictions. DIS generates 17%–21% annualized alpha on a value-weighted decile spread portfolio of relatively large stocks, well above standalone proxies on the same sample, with alphas robust to the exclusion of the most illiquid and retail-owned stocks. Our framework provides a structural explanation for why these alphas should survive standard risk adjustments. As with other disagreement proxies, alphas are concentrated in the short leg, but our model predicts this directly through a mechanism that operates independently of other channels such as aggregate sentiment and probability weighting.

Predictions extend to several additional dimensions. The model links IVOL closely to latent disagreement, and DIS absorbs IVOL's return predictability even in subsamples of smaller stocks where that predictability is strongest. The evidence extends beyond absorption. IVOL's standalone predictability is regime-dependent and friction-dependent, while DIS's is neither, identifying two specific sources of contamination in IVOL that the model's equilibrium structure appears to filter out. Together, these results provide a micro-founded, disagreement-based account of the IVOL puzzle that not only absorbs IVOL's signal but diagnoses why that signal can be unreliable.

Our framework implies that a well-specified idiosyncratic disagreement measure should predict returns independently of macro conditions. DIS is consistent with this restriction, predicting returns with consistent predictability across high and low states of eight diverse aggregate indicators, while individual proxies exhibit regime-dependence consistent with contamination by aggregate variation. This contrast helps explain why studies using different proxies across different samples can reach conflicting conclusions.

[17] Table I.8 in Internet Appendix I extends the analysis of Table 9 to additional characteristics showing further that in the absence of disagreement detected by an extant disagreement measure (IVOL, AFD, and SIR), DIS exhibits a consistent negative relationship with EXRET_{t+1}, IO, and SV, and a consistent positive relationship with KZ, OHLSON, NI-IVOL, HHI, R&D, ISS, ASC, BIDASK, and SV-INV.

Table 10
Average stock characteristics in DIS deciles.

	Low	2	3	4	5	6	7	8	9	High	H–L	*t*-stat.
DIS	1.00	1.59	2.12	2.76	3.63	4.93	7.12	11.19	20.25	67.67	66.67***	(8.64)
SIR	0.02	0.03	0.04	0.04	0.05	0.05	0.05	0.06	0.06	0.07	0.04***	(13.79)
IVOL	0.83	1.16	1.41	1.64	1.87	2.10	2.33	2.58	2.96	4.13	3.30***	(34.86)
ISKEW	0.45	0.51	0.55	0.60	0.65	0.73	0.85	1.00	1.19	1.54	1.09***	(13.08)
AFD	3.03	4.15	5.47	7.19	9.24	11.61	14.88	19.51	24.73	33.15	30.12***	(22.01)
SIZE	24.80	15.69	11.33	8.36	6.12	4.81	3.48	2.28	1.50	0.79	−24.01***	(−10.91)
BTM	0.52	0.50	0.50	0.50	0.51	0.53	0.55	0.59	0.62	0.66	0.14***	(4.97)
MOM	15.07	17.39	20.03	21.79	23.52	24.20	22.17	19.41	14.50	8.08	−6.99*	(−1.77)
BETA	0.74	0.86	0.93	0.99	1.05	1.12	1.18	1.23	1.30	1.42	0.69***	(15.25)
ILLIQ	0.01	0.01	0.02	0.03	0.04	0.05	0.07	0.11	0.21	0.46	0.45***	(5.18)
RMAX	1.60	2.06	2.39	2.68	2.97	3.28	3.58	3.91	4.43	5.94	4.34***	(26.94)

Notes: This table reports the average values of various stock characteristics across univariate, equally weighted portfolios sorted based on disagreement (DIS). The sample ranges from January 1994 to December 2022. The categories "Low" and "High" refer to the portfolios within the lowest and highest disagreement deciles, respectively. Newey and West (1987) adjusted *t*-statistics for assessing the statistical difference between the highest and lowest disagreement decile are shown in the last column. For detailed definitions of the variables, please refer to Table 1 and Appendix C.

The interaction regressions with institutional ownership provide evidence of a further distinction that disagreement depresses expected returns through two complementary channels. The individual proxies, IVOL and short interest in particular, exhibit return predictability concentrated where short-selling frictions bind and absent as institutional ownership approaches 100%, consistent with the friction-based channel of Miller (1977). In contrast, DIS exhibits significant predictability as institutional ownership approaches 100% that is insensitive to the level of frictions, consistent with the frictionless skewness channel identified by our model. That both patterns emerge from a common specification on the same sample establishes their empirical separability. The evidence that the frictionless channel operates alongside the friction-based channel provides an empirical answer to a question debated largely on theoretical grounds since Miller (1977), with implications for how the field interprets the large prior literature that conditions disagreement effects on short-sale constraints.

Our model is certainly not the last word on disagreement and there are several directions that merit future investigation. Whether the magnitude of the disagreement premium varies cross-sectionally with the ratio of prudence to risk aversion across investors remains an open empirical question. Because the ratio governs the magnitude rather than the existence of the disagreement effect, such variation would not qualitatively alter our conclusions but could provide independent evidence for the prudence mechanism. The regime-dependence diagnostic developed here has broader applicability. In any setting where a firm-level proxy is intended to capture an idiosyncratic phenomenon (e.g., information asymmetry, financial constraints, investor attention, managerial quality) the same logic applies. If the underlying phenomenon is genuinely idiosyncratic, the proxy's predictive power should be invariant to aggregate conditions under a standard factor structure. Regime-dependence of predictive power is then diagnostic of contamination by aggregate variation, not a feature of the phenomenon itself. This principle may prove useful in other contexts where researchers must distinguish idiosyncratic from aggregate signals using imperfect proxies.

CRediT authorship contribution statement

Christian L. Goulding: Formal analysis. **Campbell R. Harvey:** Formal analysis. **Hrvoje Kurtović:** Formal analysis.

Declaration of competing interest

The authors have nothing to disclose.

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Appendix A. Proofs

Proof of Proposition 1. Denote date–1 wealth for investor type i as a function of price p by $w_i(p) = w_0 + x_i(p)(\tilde{y} - p)$ so that the first-order condition of type-i investor's expected utility maximization is

$$\mathrm{E}_i[u'(w_i(p))(\tilde{y} - p)] = 0. \tag{A.1}$$

We will use date–1 wealth and (A.1) and differentiation applied twice to each to derive exact expressions for the coefficients $x_i(\mu_i)$, $x_i'(\mu_i)$, and $x_i''(\mu_i)$ of the Taylor expansion of $x_i(p)$ around $\mu_i = \mu + D\Delta_i$, namely,

$$x_i(p) = x_i(\mu_i) + x_i'(\mu_i)(p - \mu_i) + \frac{x_i''(\mu_i)}{2}(p - \mu_i)^2 + h_i(p)(p - \mu_i)^2, \tag{A.2}$$

where $h_i(p)$ is an unknown function that converges to 0 as $p \to \mu_i$ by Taylor's theorem.

First, because $w_i(\mu_i) = w_0$ for $x_i(\mu_i) = 0$, it is clear that

$$x_i(\mu_i) = 0 \tag{A.3}$$

satisfies (A.1) and this solution is unique because u is strictly concave.

Second, we differentiate both sides of (A.1) with respect to p to obtain:

$$\mathrm{E}_i[u''(w_i(p))w_i'(p)(\tilde{y} - p) - u'(w_i(p))] = 0. \tag{A.4}$$

Note that the first derivative of wealth is $w_i'(p) = x_i'(p)(\tilde{y} - p) - x_i(p)$, which evaluates to $w_i'(\mu_i) = x_i'(\mu_i)(\tilde{y} - \mu_i)$ because of (A.3). From (A.4) evaluated at $p = \mu_i$, we derive that $u''(w_0)x_i'(\mu_i)\mathrm{E}_i[(\tilde{y} - \mu_i)^2] - u'(w_0) = 0$, and therefore that

$$x_i'(\mu_i) = -\frac{1}{\mathcal{A}}\frac{1}{\sigma^2}, \tag{A.5}$$

where we have used the identity $\sigma^2 = \mathrm{E}_i[(\tilde{y} - \mu_i)^2]$ and the definition of absolute risk aversion at initial wealth, $\mathcal{A} = -\frac{u''(w_0)}{u'(w_0)}$. Note that $x_i(p)$ is locally strictly decreasing because $x_i'(\mu_i) < 0$, which follows by (A.5) and the facts that $\mathcal{A} > 0$ and $\sigma^2 > 0$.

Third, we differentiate both sides of (A.4) with respect to p to obtain:

$$\mathrm{E}_i[u'''(w_i(p))(w_i'(p))^2(\widetilde{y}-p)+u''(w_i(p))(w_i''(p)(\widetilde{y}-p)-2w_i'(p))]=0. \quad \text{(A.6)}$$

Note that the second derivative of wealth is $w_i''(p)=x_i''(p)(\widetilde{y}-p)-2x_i'(p)$, which evaluates to $w_i''(\mu_i)=x_i''(\mu_i)(\widetilde{y}-\mu_i)-2x_i'(\mu_i)$. From (A.6) evaluated at $p=\mu_i$, we derive that $u'''(w_0)(x_i'(\mu_i))^2\mathrm{E}_i[(\widetilde{y}-\mu_i)^3]+u''(w_0)x_i''(\mu_i)\mathrm{E}_i[(\widetilde{y}-\mu_i)^2]-4u''(w_0)x_i'(\mu_i)\mathrm{E}_i[\widetilde{y}-\mu_i]=0$, so

$$x_i''(\mu_i)=\frac{\mathcal{P}}{\mathcal{A}^2}\frac{s}{\sigma^3}, \quad \text{(A.7)}$$

where we have used the identities (A.5), $s\sigma^3=\mathrm{E}_i[(\widetilde{y}-\mu_i)^3]$, $\sigma^2=\mathrm{E}_i[(\widetilde{y}-\mu_i)^2]$, and $0=\mathrm{E}_i[\widetilde{y}-\mu_i]$, and the definition of absolute prudence at initial wealth, $\mathcal{P}=-\frac{u'''(w_0)}{u''(w_0)}$. For $s>0$, we have $x_i''(\mu_i)>0$, which follows by (A.7) and the facts that $\mathcal{P}>0$ and $\sigma^3>0$.

Finally, we plug in (A.3), (A.5), and (A.7) into (A.2) to obtain $x_i(p)=-\frac{1}{\mathcal{A}}\frac{1}{\sigma^2}(p-\mu_i)+\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}^2}\frac{s}{\sigma^3}(p-\mu_i)^2+h_i(p)(p-\mu_i)^2$, where $h_i(p)$ is an unknown function that converges to 0 as $p\to\mu_i$ by Taylor's theorem. □

Proof of Proposition 2. In the Taylor expansion for $x_i(p)$,

$$x_i(p)=-\frac{1}{\mathcal{A}}\frac{1}{\sigma^2}(p-\mu_i)+\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}^2}\frac{s}{\sigma^3}(p-\mu_i)^2+h_i(p)(p-\mu_i)^2, \quad \text{(A.8)}$$

note that each of the expressions $x_i'(\mu_i)=-\frac{1}{\mathcal{A}}\frac{1}{\sigma^2}$ and $x_i''(\mu_i)=\frac{\mathcal{P}}{\mathcal{A}^2}\frac{s}{\sigma^3}$ do not depend on i. Thus, for simplicity we drop the explicit reference to argument μ_i and write X_1 and X_2 to denote $x_i'(\mu_i)$ and $x_i''(\mu_i)$, respectively, to express (A.8) equivalently as

$$x_i(p)=X_1(p-\mu_i)+\frac{1}{2}X_2(p-\mu_i)^2+h_i(p)(p-\mu_i)^2. \quad \text{(A.9)}$$

Substituting $x_i(p)$ from (A.9) into the market-clearing condition $\sum_{i=1}^n\pi_i x_i(p)=0$ yields

$$X_1\sum_{i=1}^n\pi_i(p-\mu_i)+\frac{1}{2}X_2\sum_{i=1}^n\pi_i(p-\mu_i)^2+H(p)=0, \quad \text{(A.10)}$$

where we define

$$H(p)=\sum_{i=1}^n\pi_i h_i(p)(p-\mu_i)^2. \quad \text{(A.11)}$$

First, note that $\sum_{i=1}^n\pi_i(p-\mu_i)=p-\mu$ because $\widetilde{\Delta}$ has zero mean: $\sum_{i=1}^n\pi_i\Delta_i=0$. Second, note that $\sum_{i=1}^n\pi_i(p-\mu_i)^2=\sum_{i=1}^n\pi_i(p-\mu-D\Delta_i)^2=\sum_{i=1}^n\pi_i[(p-\mu)^2-2D\Delta_i(p-\mu)+D^2\Delta_i^2]=(p-\mu)^2+D^2$ because $\widetilde{\Delta}$ has zero mean and unit variance: $\sum_{i=1}^n\pi_i\Delta_i^2=1$. So, (A.10) becomes

$$X_1(p-\mu)+\frac{1}{2}X_2[(p-\mu)^2+D^2]+H(p)=0. \quad \text{(A.12)}$$

We now treat price as a function of D and use (A.12) to compute the Taylor expansion of price around $D=0$:

$$p(D)=p(0)+p'(0)D+\frac{1}{2}p''(0)D^2+h(D)D^2, \quad \text{(A.13)}$$

where $h(D)$ is an unknown function that converges to 0 as $D\to0$ by Taylor's theorem.

First, at $D=0$ it is clear that

$$p(0)=\mu \quad \text{(A.14)}$$

satisfies (A.10) because $\mu_i=\mu$ at $D=0$.

Second, we differentiate both sides of (A.12) with respect to D to obtain:

$$X_1p'+\frac{1}{2}X_2\left[2(p-\mu)p'+2D\right]+\frac{\partial H(p)}{\partial D}=0. \quad \text{(A.15)}$$

Note that $\frac{\partial H(p)}{\partial D}$ goes to zero as $D\to0$ because it is the weighted sum of terms of the form $\frac{\partial h_i(p)(p-\mu_i)^2}{\partial D}$, which each go to zero as $D\to0$. Therefore, evaluating (A.15) at $D=0$ yields

$$p'(0)=0. \quad \text{(A.16)}$$

Third, we differentiate both sides of (A.15) with respect to D to obtain:

$$X_1p''+\frac{1}{2}X_2\left[2(p')^2+2(p-\mu)p''+2\right]+\frac{\partial^2H(p)}{\partial D^2}=0. \quad \text{(A.17)}$$

Note that $\frac{\partial^2H(p)}{\partial D^2}$ goes to zero as $D\to0$ because it is the weighted sum of terms of the form $\frac{\partial^2h_i(p)(p-\mu_i)^2}{\partial D^2}$, which each go to zero as $D\to0$. Therefore, evaluating (A.17) at $D=0$ yields

$$p''(0)=\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}. \quad \text{(A.18)}$$

Therefore, by (A.13), (A.14), (A.16), and (A.18), the Taylor expansion of price around $D=0$ is given by $p(D)=\mu+\frac{1}{2}\frac{\mathcal{P}}{\mathcal{A}}\frac{s}{\sigma}D^2+h(D)D^2$, where $h(D)$ is an unknown function that converges to 0 as $D\to0$ by Taylor's theorem. □

Proof of Proposition 3. Denote $x_i(D)$ as x_i for simplicity of notation. Without loss of generality, we assume that Δ_i is distinct for each i, and relabel investor types $i=1,\ldots,n$ so that $\Delta_1<\Delta_2<\cdots<\Delta_n$. We assume that all Δ_i are nonzero. By Lemma F.1 in Internet Appendix F, $x_1<x_2<\cdots<x_n$, which means that the equilibrium demands of different type investors line up in the same order as their Δ's.

Because $\sum_{i=1}^n\pi_i x_i=0$ by market clearing, there exists some investor type $i_x<n$ such that $x_i<0$ for all $i\le i_x$ and $x_i>0$ for all $i>i_x$. Because $\mathrm{E}[\widetilde{\Delta}]=0$, there exists some investor type $i_\Delta<n$ such that $\Delta_i<0$ for all $i\le i_\Delta$ and $\Delta_i>0$ for all $i>i_\Delta$. If $i_x\le i_\Delta$, then $x_i<0$ and $\Delta_i<0$ for all $i\le i_x$. If $i_x>i_\Delta$, then $x_i>0$ and $\Delta_i>0$ for all $i>i_x$. In either case, $-\sum_{i\le i_x}\pi_i x_i=\sum_{i>i_x}\pi_i x_i$. In the first case, $\sum_{i\le i_x}\pi_i|x_i|=-\sum_{i\le i_x}\pi_i x_i$, which is increasing in D because $\Delta_i<0$ for all $i\le i_x$ by Lemma F.2 in Internet Appendix F. In the second case, $\sum_{i>i_x}\pi_i|x_i|=\sum_{i>i_x}\pi_i x_i$, which is increasing in D because $\Delta_i>0$ for all $i>i_x$, also by Lemma F.2. Either case equals exactly half of the trading volume because of the market-clearing condition. Therefore, trading volume is increasing in D. □

Appendix B. Multiple risky assets

This appendix grounds the single-asset model of Section 2 in a multi-asset foundation in which asset payoffs follow a standard factor structure. We show how the single-asset model, which characterizes how idiosyncratic investor disagreement affects the equilibrium pricing of the asset's idiosyncratic component, is an exact implication of a fully specified multi-asset equilibrium under CARA utility, and a close approximation under CRRA utility to the standard cubic Taylor expansion of Kraus and Litzenberger (1976). This foundation establishes that the single-asset model's parameters are idiosyncratic statistics (residual volatility, residual skewness, and disagreement about the residual mean) rather than total-return statistics; and delivers a key theoretical implication that an empirical measure correctly capturing idiosyncratic disagreement should be orthogonal to aggregate conditions, a prediction tested and supported in Section 5. Lastly, we discuss how our approach parallels related literature that uses independent assets or independent components of asset returns to derive separable suboptimizations (e.g., Barberis and Huang, 2008; Goulding et al., 2023b, 2025) and why our single-asset approach is particularly appropriate for disagreement research.

B.1. The multi-asset framework

Consider a market with assets that follow a factor structure, in which individual asset payoffs are the sum of a factor component (loadings on one or more factors) and an idiosyncratic component (intercept plus residual). Specifically, consider a two-period market with dates $t\in\{0,1\}$ containing $1+K+N$ assets: a risk-free asset with price and payoff normalized to one; K factors with price p_{f_k} and payoff

$\widetilde{f}_k$ at dates $t = 0$ and $t = 1$, respectively, for $k = 1, 2, \ldots, K$; and N risky assets with price p_{θ_n} and payoff

$$\widetilde{\theta}_n = \sum_{k=1}^{K} b_{n,k}\widetilde{f}_k + \widetilde{\epsilon}_n \tag{B.1}$$

at dates $t = 0$ and $t = 1$, respectively, for $n = 1, 2, \ldots, N$, where $b_{n,k}$ is asset n's loading on factor k, and $\widetilde{\epsilon}_n$ is an idiosyncratic component that is mutually independent across assets and independent of all factors, with mean μ_n, variance σ_n^2, skewness s_n, and cumulative distribution function $F_{\epsilon_n}(\epsilon) = \mathrm{P}[\widetilde{\epsilon}_n \le \epsilon]$.

The factor structure in (B.1) is intentionally general. It accommodates the CAPM, the Fama–French factor models, and any other factor structure the researcher wishes to impose. The choice of factor model governs what counts as idiosyncratic, but does not affect the form of the separation result that follows, only the numerical values of the idiosyncratic statistics.

There is a continuum of investors, indexed by $[0, 1]$, each with initial wealth w_0 and common utility function $u(w)$ satisfying $u'(w) > 0$, $u''(w) < 0$, and $u'''(w) > 0$, the same assumptions as Section 2.2. Investors trade claims on the individual assets and on the factors that underlie them. Observe that in this setup there exists a combination of trades (replicating portfolio) that replicates the idiosyncratic component of each asset, and therefore each idiosyncratic component effectively can be traded and priced. For example, a portfolio that is long an individual asset and short its factors (in quantities proportional to the loadings for that asset) generates the same payoff as the idiosyncratic component of that asset. By the law of one price, the price of this portfolio is the price of the idiosyncratic component.

The mutual independence of idiosyncratic components $\{\widetilde{\epsilon}_n\}$ is the key structural assumption. It has strong empirical support, as average cross-sectional correlations among factor-model residuals are approximately zero. It implies that the variance of a portfolio of idiosyncratic components equals the sum of their individual variances, and analogously for third central moments, both of which are used in the separation derivation below.

Moreover, it is the idiosyncratic component of each asset that investors potentially disagree about: they agree on the factor structure but disagree about each asset's idiosyncratic expected payoff. For each asset, there is a different distribution of investor beliefs regarding the asset's idiosyncratic mean (i.e., alpha). These beliefs equivalently characterize differences of opinion about individual asset expected payoffs given the factor structure. For asset n, investor disagreement is modeled exactly as in Section 2.3: investors are of I_n types with type-i_n investors believing the idiosyncratic mean is $\mu_{i_n,n} = \mu_n + D_n\Delta_{i_n,n}$, where $\widetilde{\Delta}_n$ is a zero-mean unit-variance standardized belief variable and $D_n > 0$ is the scale of disagreement for asset n. The vector $i = (i_1, i_2, \ldots, i_N)$ denotes an investor's overall type across all assets, with the relative belief variables $\{\widetilde{\Delta}_n\}$ mutually independent across assets and independent of factors.

B.2. The separation result

Under commonly used utility functions, expected utility optimization across multiple asset demands splits into separable problems of choosing the exposure to the factors and choosing the exposures to each asset's residuals. Our single-asset model of the main text is the reduction of the multi-asset problem to one of its separable asset residual models, each of which has the same form but varies in its parameters.

An investor of overall type i chooses demands $\{x_{f_k}\}$ for factors and $\{x_{\theta_n}\}$ for risky assets to maximize expected utility of terminal wealth. Terminal wealth is

$$\widetilde{w}_i = w_0 + \sum_{k=1}^{K} x_{f_k}(\widetilde{f}_k - p_{f_k}) + \sum_{n=1}^{N} x_{\theta_n}(\widetilde{\theta}_n - p_{\theta_n}). \tag{B.2}$$

By the law of one price, which holds under no-arbitrage, the price of the claim to asset n's idiosyncratic payoff $\widetilde{\epsilon}_n$ is $p_{\epsilon_n} = p_{\theta_n} - \sum_k b_{n,k}p_{f_k}$.

Substituting (B.1) into (B.2) and collecting terms, terminal wealth can be rewritten as

$$\widetilde{w}_i = w_0 + \sum_{k=1}^{K} \hat{x}_{f_k}(\widetilde{f}_k - p_{f_k}) + \sum_{n=1}^{N} \hat{x}_{\epsilon_n}(\widetilde{\epsilon}_n - p_{\epsilon_n}), \tag{B.3}$$

where $\hat{x}_{\epsilon_n} = x_{\theta_n}$ is the demand for asset n's idiosyncratic component and $\hat{x}_{f_k} = x_{f_k} + \sum_n x_{\theta_n}b_{n,k}$ is the effective demand for factor k. Eq. (B.3) rewrites the investor's wealth as the sum of payoffs from a factor portfolio and N idiosyncratic components, with the cross terms eliminated by the law of one price. See Internet Appendix G for detailed derivations.

Proposition B.1 (*Separation Under CARA*). *Under CARA utility $u(w) = -e^{-aw}$, the investor's expected utility maximization over $\{\hat{x}_{f_k}, \hat{x}_{\epsilon_n}\}$ separates exactly into $1 + N$ independent problems: one for the factor portfolio and one for each idiosyncratic component n. The optimization for idiosyncratic component n is*

$$\hat{x}_{\epsilon_n} = \arg\max_x \ \mathrm{E}_i[u(w_0 + x(\widetilde{\epsilon}_n - p_{\epsilon_n}))], \tag{B.4}$$

which has precisely the form of the single-asset optimization in (3) of Section 2.4, where $\widetilde{y} = \widetilde{\epsilon}_n$, $x = \hat{x}_{\epsilon_n}$, and $p = p_{\epsilon_n}$.

Proof. Under CARA utility, terminal wealth in (B.3) enters the utility function through the exponential. Because the idiosyncratic components $\{\widetilde{\epsilon}_n\}$ are mutually independent and independent of the factors, the expectation of the product of exponential terms equals the product of the expectations. The joint maximization therefore factors into independent maximizations for the factor portfolio and each idiosyncratic component. See Internet Appendix G for detailed derivations. □

The separation is exact under CARA because the exponential utility function transforms the product of independent payoff distributions into independent optimization problems without approximation error. Eq. (B.4) is not an approximation of the single-asset problem in Section 2.4, it is identical to it.

Proposition B.2 (*Separation Under CRRA*). *Under CRRA utility, the separation holds approximately to the cubic Taylor expansion of Kraus and Litzenberger (1976). Specifically, if investors care about the first three moments of portfolio returns, the expected utility of a type-i investor decomposes as*

$$\mathrm{E}_i[u(\widetilde{R})] \approx V_K + \sum_{n=1}^{N}\left[\hat{z}_{\epsilon_n}\mathrm{E}_i[\widetilde{r}_{\epsilon_n}] - \frac{\gamma}{2}\hat{z}_{\epsilon_n}^2\,\mathrm{Var}_i[\widetilde{r}_{\epsilon_n}] + \frac{\gamma(1+\gamma)}{6}\hat{z}_{\epsilon_n}^3\,\mathrm{E}_i[(\widetilde{r}_{\epsilon_n} - \mathrm{E}_i[\widetilde{r}_{\epsilon_n}])^3]\right], \tag{B.5}$$

where V_K collects terms involving only the factor portfolio, $\hat{z}_{\epsilon_n} = \hat{x}_{\epsilon_n}p_{\epsilon_n}/w_0$ is the portfolio weight on idiosyncratic component n, $\widetilde{r}_{\epsilon_n}$ is the net return on that component, and $\widetilde{R}$ is the gross total return on the investor's portfolio. Because the choice variable $\hat{z}_{\epsilon_n}$ appears in a separable additive term for each n, the optimization over $\hat{z}_{\epsilon_n}$ reduces to the independent maximization of the bracketed expression for asset n, which maps to cubic order to the single-asset CRRA demand of Section 2.4.

Proof. See the detailed derivation in Internet Appendix G, which applies the independence of idiosyncratic components to decompose the variance and third central moment of total portfolio return into sums of asset-level components. □

Note that the cubic approximation is standard in the asset pricing literature and supported empirically: expected returns have limited sensitivity to fourth and higher moments.

B.3. Implications of the separation

The separation results of Propositions B.1 and B.2 have three implications that carry through to the main text and the empirical analysis. These implications jointly support the theoretical framework, the measurement strategy, and the interpretation of the disagreement premium as a firm-level phenomenon distinct from aggregate phenomena.

B.3.1. From single-asset equilibrium to cross-sectional predictions

The separation result implies that each asset n in the multi-asset economy has its own idiosyncratic equilibrium, governed by its own idiosyncratic parameters, which vary across assets in the cross section. The comparative statics of the single-asset model translate directly into testable cross-sectional predictions. For example, an asset with higher idiosyncratic disagreement, all else equal, has a higher equilibrium price and lower subsequent expected return (Proposition 2). These predictions are cross-sectional in nature as they describe differences among assets at a point in time. The single-asset comparative statics are therefore not a simplification of a cross-sectional analysis but are theoretically founded.

B.3.2. Empirical construction should use idiosyncratic statistics

The separation constrains how the model's parameters must be operationalized empirically. σ must be computed from factor-model residuals. s must be based on idiosyncratic skewness (ISKEW; Boyer et al., 2010) for the same reason. Analyst forecast dispersion, which pertains to beliefs about the firm's own earnings rather than market-wide outcomes, measures the component of D related to firm-specific public information. The composite measure DIS, constructed in Section 4 from idiosyncratic components, is the empirical realization of the model's composite disagreement variable d^*. A measure constructed from systematic rather than idiosyncratic components can conflate the idiosyncratic pricing channel with factor-level variation, a distinction that guides the specific empirical choices of Section 4 as theoretically motivated rather than arbitrary.

B.3.3. Orthogonality to aggregate conditions and factor premia

The separation result implies a testable prediction about how a correctly constructed idiosyncratic disagreement measure should behave in the data. Aggregate variation such as macroeconomic uncertainty, market-wide volatility, and investor sentiment, affects all assets through the common factor structure and is absorbed into factor premia. It therefore does not enter the idiosyncratic pricing channel. This implication is exact under CARA utility (Proposition B.1) and holds to the cubic approximation under CRRA utility (Proposition B.2); since the cubic expansion is standard in the asset pricing literature and expected returns have limited sensitivity to fourth and higher moments, the orthogonality prediction applies with high accuracy under both utility specifications. If an empirical disagreement measure is correctly capturing idiosyncratic disagreement, its return predictability should be stable across aggregate macro states: the disagreement effect should be present in both high- and low-uncertainty environments, both high- and low-sentiment environments, and both high- and low-volatility environments, because none of these aggregate conditions affect the idiosyncratic channel through which the mechanism operates.

Conversely, if an empirical measure exhibits strong regime-dependence by predicting returns in some aggregate states but not others, then this behavior is diagnostic of contamination by aggregate variation that the model's idiosyncratic channel excludes. Such a measure is, at least in part, capturing a macro-regime effect rather than genuine firm-level disagreement. This prediction is tested in Section 5.4, where we show that DIS predicts returns stably across eight distinct macro state variables while IVOL and SIR predict returns only in specific regimes and AFD's predictability is less stable.

Furthermore, there is a clean distinction between idiosyncratic equilibrium pricing and variation in factor loadings, which determines exposure to factor premia. The comparative statics of Section 2 characterize how idiosyncratic characteristics generate cross-sectional variation in expected returns beyond what factor loadings predict. This is precisely the variation that Fama–MacBeth regressions with factor-based controls are designed to isolate, and it is why the empirical tests of Section 5 consistently control for beta, size, book-to-market, and momentum: these controls absorb the factor-loading channel, leaving the idiosyncratic disagreement channel identified.

The large and robust alphas documented in Section 5, which survive CAPM, Fama–French three-factor, and Carhart four-factor adjustments, align with model predictions: factor models absorb the first source of cross-sectional return variation but leave the idiosyncratic disagreement channel unaddressed, because that channel operates through asset-specific equilibrium pricing rather than through covariance with priced factors. The multi-asset framework thus not only provides a precise theoretical foundation for the single-asset approach, but also provides the structural explanation for why disagreement alphas survive standard risk adjustments.

B.4. Single risky asset analysis and cross-sectional implications

This section situates our approach within the broader literature on single-asset frameworks as tools for generating cross-sectional predictions and highlights why the single-asset approach is particularly appropriate for disagreement research.

Relationship to the literature on single-asset frameworks and cross-sectional implications

Our approach is consistent with a well-established tradition in behavioral asset pricing and market microstructure of deriving cross-sectional predictions from single-asset models via comparative statics. Kraus and Litzenberger (1976) introduce skewness into a single-asset valuation model and derive cross-sectional predictions about how distributional characteristics of assets affect their equilibrium prices. Glosten and Milgrom (1985) and Amihud and Mendelson (1986) model single assets with information asymmetry and transaction costs respectively, using comparative statics on bid–ask spreads and liquidity to derive cross-sectional liquidity premia. Mitton and Vorkink (2007) use a single representative skewed asset in a mean–variance-skewness framework to derive cross-sectional implications about skewness preference and underdiversification. Barberis and Huang (2008) study cross-sectional relationships between positive return skewness and expected returns by considering an independent skewed security added to a baseline multi-asset market. Goulding et al. (2023b) study cross-sectional relationships among skewness, noise, and expected returns in a single-asset framework. Goulding et al. (2025) study cross-sectional relationships among expected returns, skewness, and disagreement as measured by analyst forecast dispersion, using a single-asset framework.

These papers share the use of independence assumptions, either explicit or implicit, to isolate idiosyncratic characteristics so that each asset's pricing can be treated separately, mirroring a single-asset setup. We make this structure fully explicit through the formal separation results of Propositions B.1 and B.2, which derive the independence of idiosyncratic pricing from the factor structure of the multi-asset economy.

Appropriateness for disagreement research

A single-asset framework is especially well-suited to the study of idiosyncratic investor disagreement for reasons specific to the nature of the phenomenon. Disagreement, as modeled here and in the broader differences-of-opinion literature (Varian, 1985; Harris and Raviv, 1993; Kandel and Pearson, 1995; Cao and Ou-Yang, 2009) is inherently asset-specific: investors disagree about the expected payoff of a particular firm, not about aggregate market outcomes. The empirical proxies

for disagreement studied in this paper such as analyst forecast dispersion for individual firms, idiosyncratic volatility, and firm-level short interest, are all firm-specific variables. The single-asset model, in which the scale of disagreement D is a parameter of the individual asset's equilibrium, captures this asset-specificity directly. A model in which disagreement were a market-wide variable would fundamentally mischaracterize the phenomenon and predict that all assets are simultaneously more or less disagreed upon as aggregate conditions change, contradicting both the near-zero cross-correlations among firm-level disagreement proxies documented in Table 2 and the empirical finding in Section 5.4 that DIS predicts returns stably across aggregate macro conditions.

This last point connects the methodological justification for the single-asset approach to our empirical findings in a way that goes beyond precedent in the literature. The framework, the measurement, and the empirical behavior of the measure are mutually consistent in a way that provides strong joint support for the paper's central claims.

Appendix C. Empirical variable descriptions

All variables are properly lagged to be available in real time.

AFD is the standard deviation of fiscal year end analyst EPS forecasts valid in month t across analysts for stock i scaled by the average absolute EPS forecast, multiplied by 100.

Aggregate IVOL is the value-weighted average of stock-level idiosyncratic volatilities (IVOL) across all stocks in the sample in month t.

ANALYSTADJ is the number of analysts that issued a valid fiscal year forecast in month t divided by the average number of analysts per firm with valid forecasts in month t.

ASIR is the difference between a firm's actual and predicted SIR in month t. Predicted SIR is computed as the moving average SIR over the past twelve months.

BETA is the coefficient on daily market returns of the regression of daily returns on daily market returns for month t.

BIDASK is the high-low bid–ask estimator for month t following Corwin and Schultz (2012).

BTM is the ratio of book value of equity (following Davis et al., 2000) from the most recent quarter to the market value of equity which is defined as shares outstanding times the share price in month t.

CAPM-α is the intercept from the regression of monthly excess portfolio returns on a constant and excess market return for month $t+1$.

CATFIN is the systemic risk measure of Allen et al. (2012), which captures distress in the financial sector using the cross-sectional distribution of financial institution returns.

DBREADTH is the change in the ratio of mutual funds with long positions in stock i in quarter q to all mutual funds, multiplied by minus one.

DEW is the equally weighted average disagreement decile in month t based on same-month deciles of AFD, IVOL, and SIR.

DEW+ is the equally weighted average disagreement decile in month t based on same-month deciles of ASIR, BIDASK, DBREADTH, AFD, DTO, IVOL, and SUV.

DIS is the natural log of one plus our composite measure of disagreement in Eq. (15).

DPCA is the first component of a principal component analysis on AFD, IVOL, and SIR in month t.

DPCA+ is the first component of a principal component analysis on ASIR, BIDASK, DBREADTH, AFD, DTO, IVOL, and SUV in month t.

DTO is the average daily market-adjusted turnover in month t less the average daily market-adjusted turnover over the past three months.

EXRET_{t+1} is the difference between monthly raw returns and the risk-free rate for stock i in month $t+1$.

FF3-α is the intercept from the regression of monthly excess portfolio returns on a constant, the excess market return, a size factor, and a book-to-market factor in month $t+1$ following Fama and French (1993).

FFC-α is the intercept from the regression of monthly excess portfolio returns on a constant, the excess market return, a size factor, a book-to-market factor, and a momentum factor in month $t+1$ following Fama and French (1993) and Carhart (1997).

Financial Uncertainty is the financial uncertainty index of Jurado et al. (2015), based on the common variation in the unforecastable component of a large panel of financial time series at a one-month horizon.

ILLIQ averages the ratio of absolute daily returns to daily dollar trading volume, multiplied by 10^6, as in Amihud (2002).

IO is the ratio of shares held by institutional investors to total shares outstanding as in Chen et al. (2002).

ISKEW is the expected idiosyncratic skewness as defined in Boyer et al. (2010).

IVOL is the standard deviation of the daily residuals estimated from the regression of excess stock returns on a constant, the excess market return, a size factor, and a book-to-market factor at a monthly level following Ang et al. (2006), multiplied by 100.

LnSIZE is the natural logarithm of SIZE.

Macroeconomic Uncertainty is the macroeconomic uncertainty index of Jurado et al. (2015), based on the common variation in the unforecastable component of a large panel of macroeconomic indicators at a one-month horizon.

MOM is the product of monthly gross returns for month $t-12$ to $t-1$, minus 1, multiplied by 100.

PLS Sentiment Index is the partial-least-squares investor sentiment index of Kelly and Pruitt (2013), which extracts the common sentiment component from a panel of proxies using a partial least squares methodology.

R&D is research and development expenses scaled by assets.

Real Uncertainty is the real activity uncertainty index of Jurado et al. (2015), constructed analogously from a panel of real economic activity variables.

RET_{t+1} is the raw monthly return for month $t+1$, multiplied by 100.

RMAX is the average of the five highest daily returns for stock i in month t, times 100.

RVOL is the one-month ex-post return volatility based on daily data for stock i in month t, multiplied by 100.

SIR is the ratio of shares sold short to shares outstanding for stock i in month t following Chen et al. (2019).

SIZE is the product of shares outstanding and the share price of stock i in month t.

SUV is the standardized residual of a regression of trading volume on positive and negative returns over the past three months following Garfinkel (2009).

TO is the average ratio of daily trading volume to shares outstanding in month t.

VIX is the CBOE Volatility Index, which measures the implied volatility of S&P 500 index options over the next 30 days. Monthly observations correspond to the end-of-month VIX level.

Volatility-of-Volatility is the realized volatility of daily changes in the VIX over the prior calendar month, following Bekaert and Hoerova (2014).

Appendix D. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2026.104379.

Data availability

Disagreement of Disagreement (Original data) (Mendeley Data)

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